

Local PageRank Bandits for Sublinear-Time Online Link Prediction

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Abstract

Link prediction powers many real-world graph applications in which both the graph structure and user preferences evolve over time. While modern link predictors, from heuristic scores to graph neural networks, can achieve strong offline accuracy, they are typically trained on static snapshots and are costly to update at scale, making them ill-suited for online decision-making with limited feedback. We propose a sublinear-time contextual bandit framework for dynamic link prediction that couples bandit learning with local graph propagation on evolving graphs. At each round, our method assigns context-dependent scores to a small candidate set and then applies an efficient local diffusion solver to propagate these scores with sublinear work in the graph size, enabling responsive updates without touching the full graph. We further provide a regret guarantee that cleanly separates (i) statistical learning error from (ii) propagation approximation error induced by the local diffusion solver, along with a reward design tailored to link prediction. Empirically, on graphs with millions of nodes, our approach consistently improves online link prediction quality over strong bandit and graph baselines while maintaining practical runtimes.

1. Introduction

Link prediction asks a simple but consequential question: *which two nodes will connect next?* Accurate link prediction underpins many real-world systems, from friend and content recommendation in social platforms to knowledge graph completion and biological interaction discovery (Liben-Nowell & Kleinberg, 2003; Nickel et al., 2015; Lü & Zhou, 2011). Modern graphs are also *dynamic*—edges appear continuously, user interests drift, and the surrounding

topology evolves over time (Leskovec et al., 2007). In such settings, a link predictor is not only making predictions but effectively making *decisions* that change the future data it will observe.

A large literature studies link prediction as a supervised learning problem. Classical solutions rely on hand-crafted similarity scores such as common neighbors and related heuristics (Adamic & Adar, 2003; Zhou et al., 2009). Representation learning then improves flexibility via node embeddings (e.g., DeepWalk, node2vec, LINE) (Perozzi et al., 2014; Grover & Leskovec, 2016; Tang et al., 2015), and graph neural networks (GNNs) further boost accuracy using end-to-end message passing, including graph auto-encoders and subgraph-based predictors (Thomas & Welling, 2016; Zhang & Chen, 2018). Despite their success, these supervised pipelines are often mismatched with online recommendation and evolving graphs: they depend on static train/test splits, struggle to adapt to nonstationary structure, and provide limited guidance on *how to explore* when feedback is revealed only for selected candidates (Rossi et al., 2020; Cong et al., 2023). In particular, in interactive deployments, repeatedly selecting only high-confidence edges can reinforce popular nodes, while ignoring uncertain candidates may miss emerging connections.

This motivates viewing dynamic link prediction through the lens of *contextual bandits*: each round presents a serving node with a candidate set, the learner chooses one candidate (an arm), receives immediate binary feedback, and must balance exploration and exploitation over time (Li et al., 2010; Chu et al., 2011; Zhou et al., 2020). However, a bandit formulation alone is not enough for graph-scale systems. A key challenge is that ranking candidates should exploit the surrounding graph structure, yet exact graph-based propagation (e.g., Personalized PageRank) is too expensive to run at every round on large evolving networks (Tong et al., 2006).

In this paper, we propose LOCPRB (**Local PageRank Bandits**), a sequential decision-making framework for dynamic link prediction that tightly couples contextual bandit learning with *fast* graph propagation. At each round, LOCPRB constructs a sparse personalization vector from candidate-wise contextual scores and propagates it over the current graph using an ϵ -approximate Personalized PageRank (PPR) computation. Crucially, we adopt local push-style algo-

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gorithms that exploit graph locality, enabling sublinear-time propagation with provable accuracy guarantees (Andersen et al., 2006; Yoon et al., 2018a; Li et al., 2023). This design yields a practical algorithm that is responsive to graph evolution while retaining a clean regret analysis that separates contextual learning error from propagation error.

Our main contributions are threefold:

- We formulate dynamic link prediction as a contextual bandit problem on an evolving graph and introduce LOCPRB, which combines candidate-wise contextual scoring with efficient approximate PPR propagation.
- We provide regret guarantees for LOCPRB that explicitly account for both neural contextual prediction complexity and the approximation error induced by fast propagation.
- We evaluate LOCPRB on large real-world graphs and show that it improves online link prediction accuracy while substantially reducing per-round computation compared to existing bandit-based and graph-based baselines¹

2. Related Work

Contextual Bandits. Contextual bandits model sequential decision-making with side information and are central to personalized recommendation and online services (Li et al., 2010; 2011; Langford & Zhang, 2008). For linear or generalized linear rewards, ridge-regression estimators with optimism or posterior sampling achieve near-optimal regret, including LinUCB/OFUL-style methods and linear Thompson sampling (Chu et al., 2011; Abbasi-Yadkori et al., 2011a; Dani et al., 2008; Agrawal & Goyal, 2013; Abeille & Lazaric, 2017; Filippi et al., 2010; Li et al., 2017). Beyond shared parametric models, clustering and collaborative structures capture user/arm heterogeneity (Li et al., 2016; Wu et al., 2016; Gentile et al., 2014; 2017; Ban & He, 2021). Neural contextual bandits further relax linear assumptions and enable exploration in learned representations (Riquelme et al., 2018; Lu & Van Roy, 2017; Zhou et al., 2020; Zhang et al., 2021; Kassraie & Krause, 2022; Jia et al., 2022; Ban et al., 2022; 2023), while more deployment-oriented settings motivate robustness and cross-domain adaptation (Qi et al., 2024; Huang et al., 2025). Graph-structured side information has also been incorporated to exploit relational inductive biases (Qi et al., 2022; 2023; Chee et al., 2025). LocPRB is most related in spirit, but targets *dynamic link prediction* where contexts come from an evolving graph and learning must be coupled with efficient propagation.

Link Prediction Models. Link prediction is traditionally studied in supervised settings, ranging from node embeddings (DeepWalk/node2vec/LINE) (Perozzi et al., 2014; Grover & Leskovec, 2016; Tang et al., 2015) and local

¹Anonymized code is provided in the supplementary material for review and will be released publicly upon publication.

heuristics (Liben-Nowell & Kleinberg, 2003; Adamic & Adar, 2003; Barabási & Albert, 1999; Zhou et al., 2009) to GNN-based models that increase expressivity, including graph auto-encoders (Kipf & Welling, 2016; Thomas & Welling, 2016), subgraph methods such as SEAL (Zhang & Chen, 2018), and architectures combining message passing with structural signals (Yun et al., 2021; Chamberlain et al., 2022; Zhu et al., 2021; Wang et al., 2023). Recent work further improves robustness and generalization via stronger inductive biases and model combinations (Sun et al., 2024). For temporal/dynamic link prediction, temporal GNNs and continuous-time models learn from time-stamped interactions (Rossi et al., 2020; Xu et al., 2020; Wang et al., 2021; Tian et al., 2023; Cong et al., 2023; Yu et al., 2023), but are still mainly trained offline and evaluated on future snapshots. In contrast, we cast link prediction as *online sequential decision-making*, where each chosen edge reveals immediate feedback and shapes future graphs; this is most closely related to PageRank-style bandit formulations (Ban et al., 2024), while LocPRB emphasizes fast, scalable propagation on evolving graphs.

Personalized PageRank and Local Graph Algorithms.

Personalized PageRank (PPR) is classical tools for measuring node relevance in search and recommendation (Tong et al., 2006). Since exact computation is expensive on large graphs, a major theme is *local* diffusion algorithms with sublinear runtime, including local spectral/heat-kernel methods for clustering (Mahoney et al., 2012; Kloster & Gleich, 2014; Yin et al., 2017). Approximate PPR (APPR) via local push further provides degree-normalized ℓ_∞ guarantees with locality-sensitive work bounds (Andersen et al., 2006). For evolving graphs, incremental and tracking-based methods update PageRank/PPR efficiently under edge changes (Ohsaka et al., 2015; Rozenshtein & Gionis, 2016; Yoon et al., 2018a;b; Li et al., 2023; Zhang et al., 2017), and recent surveys summarize common principles (Bakar et al., 2024). LocPRB connects these ideas with contextual bandit learning by integrating fast approximate PPR computation into online link prediction on dynamic graphs.

3. Preliminaries

We consider an undirected evolving graph sequence $\{\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)\}_{t=0}^T$ over a fixed node set $\mathcal{V}$ with $|\mathcal{V}| = n$. The initial graph is $\mathcal{G}_0 = (\mathcal{V}, \mathcal{E}_0)$. Each node $v_i \in \mathcal{V}$ is associated with a feature vector $\mathbf{x}_{0,i} \in \mathbb{R}^d$. We study link prediction on this evolving graph under a contextual bandit protocol for T rounds, and use $[T] = \{1, 2, \dots, T\}$ to index decision rounds. For each $t \in [T]$, let $\mathbf{A}_t = [a_{uv}^{(t)}] \in \{0, 1\}^{n \times n}$ denote the adjacency matrix of $\mathcal{G}_t$, where $a_{uv}^{(t)} = 1$ iff $(u, v) \in \mathcal{E}_t$. For any node $v \in \mathcal{V}$, its neighborhood and degree at time t are $\mathcal{N}_t(v) \triangleq \{u \in \mathcal{V} : a_{uv}^{(t)} = 1\}$ and $d_v^{(t)} \triangleq |\mathcal{N}_t(v)|$, respectively. The degree matrix $\mathbf{D}_t$ is diag-

onal with entries $D_{vv}^{(t)} = d_v^{(t)}$. For any node subset $\mathcal{S} \subseteq \mathcal{V}$, we define its volume at time t as $\text{vol}_t(\mathcal{S}) \triangleq \sum_{v \in \mathcal{S}} d_v^{(t)}$, and in particular $\text{vol}_t(\mathcal{V}) = 2|\mathcal{E}_t|$. For a vector $\mathbf{x} \in \mathbb{R}^n$, we write $\text{supp}(\mathbf{x}) \triangleq \{v \in [n] : x_v \neq 0\}$ for its support, and use $\mathbf{x}(u)$ to denote the entry indexed by node u .

3.1. Bandit Protocol

At each round t , the learner observes the current graph $\mathcal{G}_{t-1}$, a serving node $v_t \in \mathcal{V}$, and a candidate set $\mathcal{V}_t = \{v_{t,1}, \dots, v_{t,k}\} \subseteq \mathcal{V}$ with contexts $\mathcal{X}_t = \{\mathbf{x}_{t,1}, \dots, \mathbf{x}_{t,k}\}$. The learner selects an index $i \in [k]$ and receives binary feedback

$$r_{t,i} = \begin{cases} 1, & \text{if edge } (v_t, v_{t,i}) \text{ is formed,} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The reward is generated according to a conditional distribution $\mathcal{D}_{\mathcal{Y}|\mathbf{x}_{t,i}}$ over $\mathcal{Y} = \{0, 1\}$. Let $i_t^* = \arg \max_{v_{t,i} \in \mathcal{V}_t} \mathbb{P}(r_{t,i} = 1 \mid \mathbf{x}_{t,i})$ denote the optimal arm and $\hat{i}_t$ the arm chosen by the learner. The cumulative pseudo-regret is

$$\begin{aligned} \mathbf{R}_T &= \sum_{t=1}^T \left(\mathbb{E}_{r_{t,i_t^*} \sim \mathcal{D}_{\mathcal{Y}|\mathbf{x}_{t,i_t^*}}} [r_{t,i_t^*}] - \mathbb{E}_{r_{t,\hat{i}_t} \sim \mathcal{D}_{\mathcal{Y}|\mathbf{x}_{t,\hat{i}_t}}} [r_{t,\hat{i}_t}] \right) \\ &= \sum_{t=1}^T \left(\mathbb{P}(r_{t,i_t^*} = 1 \mid \mathbf{x}_{t,i_t^*}) - \mathbb{P}(r_{t,\hat{i}_t} = 1 \mid \mathbf{x}_{t,\hat{i}_t}) \right), \end{aligned}$$

and the goal is to minimize $\mathbf{R}_T$.

3.2. Personalized PageRank

To incorporate graph structure into decision-making, we use Personalized PageRank (PPR). Given a v_t , a personalization vector $\mathbf{s}_t \in \mathbb{R}^n$, and a damping factor $\alpha \in (0, 1)$, the PPR vector $\boldsymbol{\pi}_t \in \mathbb{R}^n$ on $\mathcal{G}_{t-1}$ is the unique solution to

$$\boldsymbol{\pi}_t = \alpha \mathbf{P}_{t-1} \boldsymbol{\pi}_t + (1 - \alpha) \mathbf{s}_t, \quad (2)$$

where the random-walk transition matrix is $\mathbf{P}_{t-1} \triangleq \mathbf{D}_{t-1}^{-1} \mathbf{A}_{t-1}$. Without loss of generality, we assume $\|\mathbf{s}_t\|_1 = 1$; otherwise one can rescale $\mathbf{s}_t \leftarrow \mathbf{s}_t / \|\mathbf{s}_t\|_1$ and the resulting $\boldsymbol{\pi}_t$ scales accordingly. Equivalently, defining $\mathbf{M}_{t-1} \triangleq \mathbf{I} - \alpha \mathbf{P}_{t-1}$, we can write

$$\mathbf{M}_{t-1} \boldsymbol{\pi}_t = (1 - \alpha) \mathbf{s}_t, \quad (3)$$

and hence $\boldsymbol{\pi}_t = (1 - \alpha) \mathbf{M}_{t-1}^{-1} \mathbf{s}_t$.

We quantify approximation error using a degree-normalized ℓ_∞ norm. For any $\mathbf{x} \in \mathbb{R}^n$, define

$$\|\mathbf{x}\|_{\infty, \mathbf{D}_{t-1}} \triangleq \max_{u \in \mathcal{V}} \frac{|\mathbf{x}(u)|}{d_u^{(t-1)}}.$$

Given $\epsilon > 0$, we say that $\hat{\boldsymbol{\pi}}_t$ is an ϵ -approximate PPR vector if

$$\begin{aligned} \hat{\boldsymbol{\pi}}_t &\in \mathcal{P}_t(\epsilon, \alpha, \mathbf{s}_t, \mathcal{G}_{t-1}) \\ &\triangleq \{\boldsymbol{\pi} : \|(\boldsymbol{\pi} - \boldsymbol{\pi}_t)\|_{\infty, \mathbf{D}_{t-1}} \leq \epsilon\}. \end{aligned} \quad (4)$$

4. Fast PageRank Bandits

This section presents our main algorithmic framework and theoretical guarantees. We begin by formulating dynamic link prediction as a contextual bandit problem and introducing LOCPRB. We then provide a rigorous regret analysis establishing regret bounds for our approach. Finally, we discuss potential improvements and connections to related graph-based learning problems.

4.1. LocPRB Scoring and Decision Rule

Algorithm 1 summarizes LOCPRB. At each round, the learner first evaluates candidates using contextual signals and then propagates these scores over $\mathcal{G}_{t-1}$ via PPR to obtain graph-aware relevance scores. The key design choice is to perform this propagation efficiently: rather than solving (2) exactly, we compute an ϵ -approximate PPR vector using a local push procedure, which exploits the sparsity of the personalization vector and the temporal correlation between consecutive rounds.

Contextual network: exploitation and exploration We model exploitation with a neural predictor $f_1(\cdot; \theta^1)$ that maps a candidate context to its estimated reward. At round t , given θ_{t-1}^1 learned from past feedback, each candidate $v_{t,i} \in \mathcal{V}_t$ receives the exploitation score

$$\mu_{t,i} \triangleq f_1(\mathbf{x}_{t,i}; \theta_{t-1}^1), \quad i \in [k]. \quad (5)$$

After selecting $\hat{i}$ and observing $r_{t,\hat{i}} \in \{0, 1\}$, we update θ^1 by SGD on the squared loss

$$\mathcal{L}_1(\theta_{t-1}^1) \triangleq \frac{1}{2} (\mu_{t,\hat{i}} - r_{t,\hat{i}})^2, \quad (6)$$

yielding θ_t^1 for the next round.

To guide exploration, we learn an auxiliary network $f_2(\cdot; \theta^2)$ that predicts the potential improvement beyond the current exploitation estimate. Following gradient-based exploration, we represent candidate i by

$$\phi(\mathbf{x}_{t,i}) \triangleq \nabla_{\theta^1} f_1(\mathbf{x}_{t,i}; \theta_{t-1}^1), \quad (7)$$

and define the exploration bonus

$$b_{t,i} \triangleq f_2(\phi(\mathbf{x}_{t,i}); \theta_{t-1}^2), \quad i \in [k]. \quad (8)$$

After observing $r_{t,\hat{i}}$, the residual reward $r_{t,\hat{i}} - \mu_{t,\hat{i}}$ serves as the regression target for f_2 . We update θ^2 via SGD with

$$\mathcal{L}_2(\theta_{t-1}^2) \triangleq \frac{1}{2} (b_{t,\hat{i}} - (r_{t,\hat{i}} - \mu_{t,\hat{i}}))^2, \quad (9)$$

resulting in θ_t^2 .

Algorithm 1 LOCPRB($\epsilon, \alpha, \eta_1, \eta_2, T, \mathcal{G}_0, f_1, f_2, \mathcal{M}$)

```

1: Initialize parameters  $\theta_0^1, \theta_0^2$  and graph  $\mathcal{G}_0$ 
2: for  $t = 1, 2, \dots, T$  do
3:   Observe  $\mathcal{G}_{t-1}$ , serving node  $v_t$ , candidate set  $\mathcal{V}_t = \{v_{t,i}\}_{i=1}^k$ , and contexts  $\mathcal{X}_t = \{\mathbf{x}_{t,i}\}_{i=1}^k$ 
4:   Initialize  $\mathbf{s}_t \leftarrow \mathbf{0} \in \mathbb{R}^n$ 
5:   for  $i = 1, \dots, k$  do
6:      $\mu_{t,i} \leftarrow f_1(\mathbf{x}_{t,i}; \theta_{t-1}^1)$ 
7:      $\phi_{t,i} \leftarrow \nabla_{\theta^1} f_1(\mathbf{x}_{t,i}; \theta_{t-1}^1)$ 
8:      $b_{t,i} \leftarrow f_2(\phi_{t,i}; \theta_{t-1}^2)$ 
9:      $\mathbf{s}_t[v_{t,i}] \leftarrow \mu_{t,i} + b_{t,i}$ 
10:  end for
11:  Compute  $\hat{\pi}_t \in \mathcal{P}_t(\epsilon, \alpha, \mathbf{s}_t, \mathcal{G}_{t-1})$  by approximately solving (2) using DYN-APPR
12:  Select  $\hat{i} \in \arg \max_{i \in [k]} \hat{\pi}_t[v_{t,i}]$  and observe reward  $r_{t,\hat{i}} \in \{0, 1\}$ 
13:  if  $r_{t,\hat{i}} = 1$  then
14:     $\mathcal{G}_t \leftarrow (\mathcal{V}, \mathcal{E}_{t-1} \cup \{(v_t, v_{t,\hat{i}})\})$ 
15:  else
16:     $\mathcal{G}_t \leftarrow \mathcal{G}_{t-1}$ 
17:  end if
18:   $\theta_t^1 \leftarrow \theta_{t-1}^1 - \eta_1 \nabla_{\theta^1} \mathcal{L}_1(\theta_{t-1}^1)$ 
19:   $\theta_t^2 \leftarrow \theta_{t-1}^2 - \eta_2 \nabla_{\theta^2} \mathcal{L}_2(\theta_{t-1}^2)$ 
20: end for

```

Context-to-graph propagation via PPR. We fuse contextual learning with graph structure using a sparse personalization vector $\mathbf{s}_t \in \mathbb{R}^n$. Since only candidates in $\mathcal{V}_t$ are directly comparable at round t , we inject mass exclusively on this set:

$$\mathbf{s}_t[v] \triangleq \begin{cases} \mu_{t,i} + b_{t,i}, & \text{if } v = v_{t,i} \in \mathcal{V}_t, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Given $\alpha \in (0, 1)$, we compute the propagated scores by approximately solving (2), producing $\hat{\pi}_t$. Here we do not normalize $\mathbf{s}_t$ since scaling only rescales propagated scores. We adjust ϵ accordingly. The learner then selects the candidate with the largest propagated value,

$$\hat{i} \in \arg \max_{i \in [k]} \hat{\pi}_t[v_{t,i}]. \quad (11)$$

Finally, after receiving feedback, we update the graph to obtain $\mathcal{G}_t$ (e.g., adding $(v_t, v_{t,\hat{i}})$ when $r_{t,\hat{i}} = 1$).

4.2. Fast PPR Solver on Evolving Graphs

A key bottleneck of PRB-style methods is repeatedly solving the PPR linear system (2) on large sparse graphs. LOCPRB accelerates this step by exploiting (i) the sparsity of the personalization vector $\mathbf{s}_t$ (by construction $\text{supp}(\mathbf{s}_t) \subseteq \mathcal{V}_t$), and (ii) temporal correlation across rounds, since $\mathcal{G}_{t-1}$ and $\mathcal{G}_{t-2}$ typically differ by a small number of edge updates (e.g., a

Algorithm 2 APPR($\alpha, \epsilon, \mathbf{s}_t, \mathcal{G}_{t-1}, \mathbf{p}, \mathbf{r}$)

```

1: // If not provided, initialize  $\mathbf{p} \leftarrow \mathbf{0}$  and  $\mathbf{r} \leftarrow \mathbf{s}_t$ .
2: while  $\exists u \in \mathcal{V}$  such that  $|\mathbf{r}(u)| \geq \epsilon d_u^{(t-1)}$  do
3:    $(\mathbf{p}, \mathbf{r}) \leftarrow \text{PUSH}(u, \alpha, \mathbf{p}, \mathbf{r})$ 
4: end while
5: return  $\mathbf{p}$ 

6: Procedure PUSH( $u, \alpha, \mathbf{p}, \mathbf{r}$ )
7:  $\xi \leftarrow \mathbf{r}(u)$ 
8:  $\mathbf{p}(u) \leftarrow \mathbf{p}(u) + (1 - \alpha)\xi$ 
9:  $\mathbf{r}(u) \leftarrow 0$ 
10: for each  $v \in \mathcal{N}_{t-1}(u)$  do
11:    $\mathbf{r}(v) \leftarrow \mathbf{r}(v) + \alpha \xi / d_u^{(t-1)}$ 
12: end for
13: return  $(\mathbf{p}, \mathbf{r})$ 

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single edge insertion). We adopt the classical local Approximate Personalized PageRank (APPR) push procedure (Andersen et al., 2006) as the PPR solver in Algorithm 1, and further leverage its efficient update behavior under small graph changes as studied in dynamic APPR variants (Zhang et al., 2016). For clarity of exposition, we defer all proofs of theorems and lemmas in this subsection to Appendix B.

APPR via local pushes. Algorithm 2 computes a *local* approximation of the PPR solution by repeatedly applying a PUSH operation to nodes whose residual mass is large relative to their degrees. Starting from $\mathbf{p} \leftarrow \mathbf{0}$ and $\mathbf{r} \leftarrow \mathbf{s}_t$, APPR iteratively selects an *active* node u satisfying $|\mathbf{r}(u)| \geq \epsilon d_u^{(t-1)}$ and performs a local redistribution step that updates the estimate and propagates the remaining mass to the neighbors of u .

Concretely, one PUSH commits a $(1 - \alpha)$ -fraction of the residual at u into the current estimate, and redistributes the remaining α -fraction to $\mathcal{N}_{t-1}(u)$ according to the random-walk transition rule. In vector form, the update can be written as

$$\begin{aligned} \mathbf{p} &\leftarrow \mathbf{p} + (1 - \alpha) \mathbf{r}(u) \mathbf{e}_u, \\ \mathbf{r} &\leftarrow \mathbf{r} - \mathbf{r}(u) \mathbf{e}_u + \alpha \mathbf{r}(u) \mathbf{P}_{t-1} \mathbf{e}_u, \end{aligned} \quad (12)$$

where $\mathbf{e}_u$ denotes the indicator vector of node u . The procedure terminates when no active nodes remain, i.e., $\|\mathbf{r}\|_{\infty, \mathcal{D}_{t-1}} \leq \epsilon$, and returns $\mathbf{p}$ as an ϵ -approximate PPR vector.

A key property of the local PUSH operation is that it preserves the linear-system invariant

$$\mathbf{M}_{t-1} \mathbf{p} + (1 - \alpha) \mathbf{r} = (1 - \alpha) \mathbf{s}_t. \quad (13)$$

As a result, the residual $\mathbf{r}$ provides a direct certificate of how well $\mathbf{p}$ solves the PPR system (2). In particular, bounding $\|\mathbf{r}\|_{\infty, \mathcal{D}_{t-1}}$ immediately translates into a degree-normalized ℓ_∞ accuracy guarantee for the resulting PPR estimate.

Beyond correctness, APPR is attractive for large graphs because its computation is inherently local: with the standard activity rule, it concentrates work on nodes near the injected mass instead of traversing the entire graph. The following result summarizes the key properties of APPR that we use in our analysis.

Theorem 4.1 (APPR termination, accuracy, and locality). *Run APPR (Algorithm 2) on $\mathcal{G}_{t-1}$ with threshold ϵ and activity rule $|\mathbf{r}(u)| \geq \epsilon d_u^{(t-1)}$. Then:*

- **(Termination)** *The algorithm terminates after finitely many PUSH operations.*
- **(Accuracy)** *Upon termination, $\|\mathbf{r}\|_{\infty, D_{t-1}} \leq \epsilon$ and the returned estimate $\hat{\pi}_t := \mathbf{p}$ satisfies (4).*
- **(Locality)** *The total processed degree and explored volume satisfy*

$$\text{work}_t \text{ and } \text{vol}_{t-1}(\text{supp}(\hat{\pi}_t)) \leq \frac{1}{(1-\alpha)\epsilon}. \quad (14)$$

Theorem 4.1 characterizes APPR when computing $\hat{\pi}_t$ from scratch on $\mathcal{G}_{t-1}$. In LOCPRB, however, the graph evolves slowly across rounds (at most one edge insertion per step), so the PPR solution typically changes only locally. This motivates a dynamic variant that reuses the previous APPR state: we apply a constant-time local correction after each insertion and then invoke APPR only to restore the stopping condition $\|\mathbf{r}\|_{\infty, D} \leq \epsilon$, yielding substantially smaller work on evolving graphs.

APPR on Dynamic Graphs. Suppose $\mathcal{G}_{t-1}$ is obtained from $\mathcal{G}_{t-2}$ by inserting a single undirected edge (a, b) . Since only the neighborhoods and degrees of the endpoints change, the random-walk operator P_{t-1} (and hence M_{t-1}) is affected only locally. This locality motivates a dynamic update scheme that maintains the APPR state across rounds. Concretely, we carry over the previous pair $(\mathbf{p}, \mathbf{r})$ and apply a constant-time INSERTUPDATE correction so that the linear-system invariant (13) remains valid under the updated graph. After this correction, we invoke APPR to further reduce the residual below the stopping threshold, i.e., $\|\mathbf{r}\|_{\infty, D_{t-1}} \leq \epsilon$, and obtain an ϵ -approximate PPR vector on $\mathcal{G}_{t-1}$. The complete procedure is summarized in Algorithm 3.

We measure the *work* of an APPR call as the total processed degree over all executed PUSH operations, consistent with Theorem 4.1. Let work_t denote the work incurred by DYN-APPR (Algorithm 3) at update t . In the LOCPRB protocol, each update inserts at most one edge, so INSERTUPDATE performs only constant-time local corrections before calling APPR. This yields an amortized bound on the cumulative push work across T updates, formalized in Theorem 4.2.

Theorem 4.2 (DYN-APPR: invariant preservation and total push work). *Consider a graph sequence $\{\mathcal{G}_t\}_{t=1}^T$ where each $\mathcal{G}_t$ is obtained from $\mathcal{G}_{t-1}$ by inserting at most one*

Algorithm 3 DYN-APPR($\alpha, \epsilon, \mathbf{s}_t, \mathcal{G}_{t-1}$)

- 1: **Assume** $\mathcal{G}_{t-1}$ is obtained from $\mathcal{G}_{t-2}$ by inserting one undirected edge (a, b) .
 - 2: $\mathbf{p} \leftarrow \hat{\pi}_{t-1}$ and $\mathbf{r} \leftarrow \mathbf{r}_{t-1}$.
 - 3: $(\mathbf{p}, \mathbf{r}) \leftarrow \text{INSERTUPDATE}(a, b, \alpha, \mathbf{p}, \mathbf{r}, \mathcal{G}_{t-1})$.
 - 4: $\hat{\pi}_t \leftarrow \text{APPR}(\alpha, \epsilon, \mathbf{s}_t, \mathcal{G}_{t-1}, \mathbf{p}, \mathbf{r})$.
 - 5: **return** $\hat{\pi}_t$
 - 6: **Procedure** INSERTUPDATE($a, b, \alpha, \mathbf{p}, \mathbf{r}, \mathcal{G}$)
 - 7: INSERTONEDIR($a, b, \alpha, \mathbf{p}, \mathbf{r}, \mathcal{G}$)
 - 8: INSERTONEDIR($b, a, \alpha, \mathbf{p}, \mathbf{r}, \mathcal{G}$)
 - 9: **return** $(\mathbf{p}, \mathbf{r})$
 - 10: **Procedure** INSERTONEDIR($u, v, \alpha, \mathbf{p}, \mathbf{r}, \mathcal{G}$)
 - 11: // d_u denotes the new degree of u in $\mathcal{G}$ after insertion
 - 12: $\mathbf{p}(u) \leftarrow \mathbf{p}(u) \cdot \frac{d_u}{d_u-1}$
 - 13: $\Delta \leftarrow \frac{1}{\alpha} \cdot \frac{\mathbf{p}(u)}{d_u}$
 - 14: $\mathbf{r}(u) \leftarrow \mathbf{r}(u) - \Delta$
 - 15: $\mathbf{r}(v) \leftarrow \mathbf{r}(v) + (1-\alpha)\Delta$
 - 16: **return** $(\mathbf{p}, \mathbf{r})$
-

undirected edge. For each update t , run DYN-APPR (Algorithm 3) with $\mathbf{s}_t$. Let $(\mathbf{p}, \mathbf{r})$ denote the maintained estimate-residual state after the INSERTUPDATE step and before the subsequent APPR call, and let work_t be the total processed degree of all PUSH operations executed by that APPR call.

- **(Invariant preservation)** *The INSERTUPDATE step produces a state $(\mathbf{p}, \mathbf{r})$ satisfying the linear-system invariant (13)*
- **(Total push work bound)** *If each inserted edge has a uniformly random endpoint in $\mathcal{V}$, then*

$$\mathbb{E} \left[\sum_{t=1}^T \text{work}_t \right] = \mathcal{O} \left(\frac{1}{(1-\alpha)\epsilon} + \frac{T}{(1-\alpha)^2} + \frac{T}{n(1-\alpha)^2\epsilon} \right). \quad (15)$$

In particular, for constant α and $\epsilon = \Omega(1/n)$,

$$\mathbb{E} \left[\sum_{t=1}^T \text{work}_t \right] = \mathcal{O} \left(\frac{T}{(1-\alpha)^2} + \frac{1}{(1-\alpha)\epsilon} \right).$$

4.3. Regret Analysis

We analyze how approximate PPR propagation impacts the pseudo-regret. At each round t , each candidate $v_{t,i} \in \mathcal{V}_t$ is associated with a context $\mathbf{x}_{t,i} \in \mathbb{R}^d$ and an expected reward determined by an unknown function $\mathbf{y}(\cdot)$, i.e., $\mathbb{E}[r_{t,i} | \mathbf{x}_{t,i}] = \mathbf{y}(\mathbf{x}_{t,i})$. Define the (unknown) sparse expected-reward vector $\mathbf{y}_t \in \mathbb{R}^n$ supported on $\mathcal{V}_t$ by $\mathbf{y}_t[v_{t,i}] = \mathbb{E}[r_{t,i} | \mathbf{x}_{t,i}]$ and $\mathbf{y}_t[v] = 0$ for $v \notin \mathcal{V}_t$.

LOCPRB assigns each candidate an optimistic contextual score $\mu_{t,i} + b_{t,i}$, where $\mu_{t,i}$ is produced by the exploitation network and $b_{t,i}$ is an exploration bonus. We lift these scores

to node-indexed sparse vectors $\mu_t, \mathbf{b}_t \in \mathbb{R}^n$ supported on $\mathcal{V}_t$, with $\mu_t[v_{t,i}] = \mu_{t,i}$ and $\mathbf{b}_t[v_{t,i}] = b_{t,i}$, and set the injection vector for propagation to $\mathbf{s}_t = \mu_t + \mathbf{b}_t$ (cf. (10)).

To define the benchmark induced by true rewards, let $\pi_t^* \in \mathbb{R}^n$ be the unique PPR solution to

$$\pi_t^* = \alpha \mathbf{P}_{t-1} \pi_t^* + (1 - \alpha) \mathbf{y}_t. \quad (16)$$

We view π_t^* as the oracle propagated score vector. In contrast, LOCPRB computes an approximation $\hat{\pi}_t \in \mathbb{R}^n(\epsilon, \alpha, \mathbf{s}_t, \mathcal{G}_{t-1})$ and selects

$$\hat{i}_t \in \arg \max_{i \in [k]} \hat{\pi}_t[v_{t,i}], \quad i_t^* \in \arg \max_{i \in [k]} \pi_t^*[v_{t,i}], \quad (17)$$

which yields the pseudo-regret

$$\mathbf{R}_T = \sum_{t=1}^T \left(\pi_t^*[v_{t,i_t^*}] - \pi_t^*[v_{t,\hat{i}_t}] \right). \quad (18)$$

As is standard in contextual bandits, we assume normalized contexts $\|\mathbf{x}_{t,i}\|_2 = 1$ for all $t \in [T]$, $i \in [k]$. We consider a fully-connected ReLU network with depth $L \geq 2$ and width m :

$$f(\mathbf{x}_{t,i}; \theta) = \mathbf{W}_L \sigma(\mathbf{W}_{L-1} \sigma(\cdots \sigma(\mathbf{W}_1 \mathbf{x}_{t,i}) \cdots)), \quad (19)$$

where $\mathbf{W}_1 \in \mathbb{R}^{m \times d}$, $\mathbf{W}_\ell \in \mathbb{R}^{m \times m}$ for $2 \leq \ell \leq L-1$, and $\mathbf{W}_L \in \mathbb{R}^{1 \times m}$, and we stack parameters as $\theta = [\text{vec}(\mathbf{W}_1)^\top, \dots, \text{vec}(\mathbf{W}_L)^\top]^\top \in \mathbb{R}^p$. We adopt the standard random initialization: for $1 \leq \ell \leq L-1$, entries of $\mathbf{W}_\ell$ are i.i.d. $\mathcal{N}(0, 2/m)$, and entries of $\mathbf{W}_L$ are i.i.d. $\mathcal{N}(0, 1/m)$. The networks f_1 and f_2 in Algorithm 1 follow the same architecture and parameterization. We also define the reward vector $\mathbf{y} = [\mathbf{y}(\mathbf{x}_{t,i}) : t \in [T], i \in [k]] \in \mathbb{R}^{Tk}$.

Our analysis builds on over-parameterized neural contextual bandits through the induced depth- L Neural Tangent Kernel (NTK) matrix $\mathbf{H} \in \mathbb{R}^{Tk \times Tk}$ defined on $\{\mathbf{x}_{t,i} : t \in [T], i \in [k]\}$ (see Appendix C). The statistical term in the regret bound depends on the conditioning of $\mathbf{H}$.

Assumption 4.3 (NTK non-degeneracy). There exists $\lambda_0 > 0$ such that $\mathbf{H} \succeq \lambda_0 \mathbf{I}$.

Assumption 4.3 is standard in the neural bandit literature and ensures that the NTK regression problem is well-conditioned. We now state the regret guarantee for LOCPRB.

Theorem 4.4 (Regret of LOCPRB). *Given the number of rounds T , for any $\alpha, \delta \in (0, 1)$, suppose the network width satisfies $m \geq \tilde{\Omega}(\text{poly}(T, L)) \cdot k \log(1/\delta)$, and the step sizes are chosen as $\eta_1 = \eta_2 = \frac{T^3}{\sqrt{m}}$.*

Assume that at each round t , APPR returns an ϵ -approximate PPR vector $\hat{\pi}_t$ in the sense of (4), where the

tolerance satisfies $\epsilon = \mathcal{O}(1/\sqrt{T})$. Further assume that the evolving graph has bounded maximum degree, i.e.,

$$\max_{v \in \mathcal{V}} d_v^{(t-1)} \leq d_{\max} \quad \text{for all } t \in [T].$$

If Assumption 4.3 holds, then with probability at least $1 - \delta$ over the random initialization, Algorithm 1 achieves the regret bound

$$\mathbf{R}_T \leq \tilde{\mathcal{O}}\left(\sqrt{T} \left(d_{\max} + \sqrt{\zeta k \max(\zeta, S^2)}\right)\right), \quad (20)$$

where

$$\zeta \triangleq \frac{\log \det(\mathbf{I} + \mathbf{H})}{\log(1 + Tk)} \quad \text{and} \quad S \triangleq \sqrt{\mathbf{y}^\top \mathbf{H}^{-1} \mathbf{y}}.$$

Proof sketch. We outline the main steps and highlight how approximate propagation affects the regret decomposition. Per round, we bound the regret by (i) a stability term $\|\mathbf{y}_t - \mathbf{s}_t\|_2$ (oracle PPR versus PPR under optimistic injection) and (ii) an approximation term from inserting $\hat{\pi}_t$, which satisfies $\Delta_t \leq \epsilon d_{\max}$ by the degree-normalized APPR guarantee. Summing over t and applying standard neural-bandit squared-error bounds yields the result; setting $\epsilon = \mathcal{O}(1/\sqrt{T})$ makes the propagation error contribute at most $\mathcal{O}(d_{\max} \sqrt{T})$. See Appendix D for details.

Sublinear regret in special cases. The bound in Theorem 4.4 depends on $S = \sqrt{\mathbf{y}^\top \mathbf{H}^{-1} \mathbf{y}}$. Below are two benign regimes where S is small.

- **NTK-aligned rewards.** Assume $\mathbf{y}$ lies in an r -dimensional subspace such that $\mathbf{H} \succeq \lambda_{\text{big}} \mathbf{P}$ and $\mathbf{P} \mathbf{y} = \mathbf{y}$ for a rank- r projector $\mathbf{P}$. Then $S^2 \leq \|\mathbf{y}\|_2^2 / \lambda_{\text{big}}$. If $\|\mathbf{y}\|_2^2 = \mathcal{O}(T)$ and $\lambda_{\text{big}} = \Omega(T)$, we have $S = \mathcal{O}(1)$ and $\mathbf{R}_T = \tilde{\mathcal{O}}(\sqrt{T})$ when $\zeta, k = \mathcal{O}(1)$.
- **Rare-link rewards.** Suppose $\mathbf{y}(\mathbf{x}_{t,i}) \leq \rho_t$ with $\rho_t \leq c/t$ for all $i \in [k]$. Then $\|\mathbf{y}\|_2^2 \leq k \sum_{t=1}^T \rho_t^2 = \mathcal{O}(k)$ and thus $S^2 \leq \|\mathbf{y}\|_2^2 / \lambda_0 = \mathcal{O}(k)$ under Assumption 4.3. For $k = \mathcal{O}(1)$, Theorem 4.4 yields $\mathbf{R}_T = \tilde{\mathcal{O}}(\sqrt{T})$.

4.4. Generalization

LocPRB currently uses Personalized PageRank (PPR) to propagate candidate-wise contextual scores over the evolving graph. More broadly, many graph propagation operators used in learning and clustering can be written as *graph diffusion equations* (GDEs), i.e., generalized PageRank operators that mix random-walk lengths with a nonnegative coefficient sequence (Chung, 2007; Gleich, 2015). This section shows that the LocPRB bandit-propagation coupling extends directly to this broader family, and in particular to *heat-kernel* diffusion.

General graph diffusion operator. Let the random-walk transition matrix on $\mathcal{G}_{t-1}$ be $\mathbf{P}_{t-1} = \mathbf{D}_{t-1}^{-1} \mathbf{A}_{t-1}$. Given coefficients $\{c_k\}_{k \geq 0}$ with $c_k \geq 0$ and $\sum_{k=0}^{\infty} c_k < \infty$, define

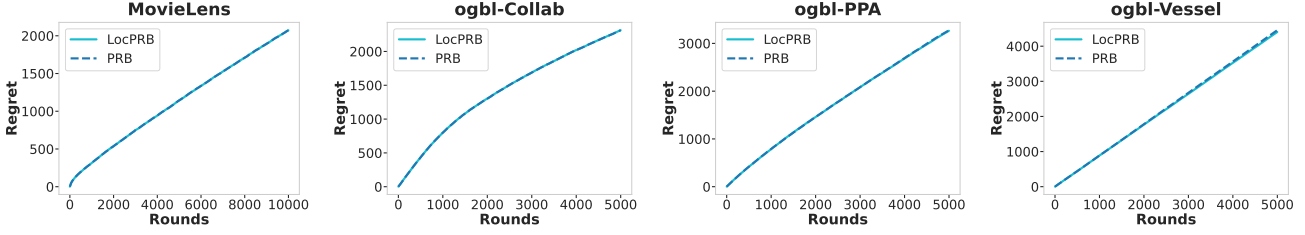


Figure 1. Regret comparison of PRB and LocPRB on online link prediction datasets (average of 10 runs, detailed in Table 1).

	MovieLens		ogbl-Collab		ogbl-PPA		ogbl-Vessel	
Method	Regret	Time	Regret	Time	Regret	Time	Regret	Time
PRB	2054±56.9	7636±103.4	2199±114.5	29952±483.3	3276±137.3	56877±6710.3	4455±28.5	15111±714.2
LocPRB	2054±56.9	4247±73.9	2194±129.5	1676±115.5	3264±155.5	3869±374.3	4404±37.4	1276±87.4

Table 1. Cumulative regret (Regret, mean ± std) and total running time (Time, mean ± std, in seconds) of PRB and LocPRB on online link prediction.

the propagated score vector and its generalized diffusion operator as follows

$$\mathbf{f}_t^{(c)} \triangleq \mathbf{K}_{t-1}^{(c)} \mathbf{s}_t, \text{ with } \mathbf{K}_{t-1}^{(c)} \triangleq \sum_{k=0}^{\infty} c_k \mathbf{P}_{t-1}^k, \quad (21)$$

where for any injected signal $\mathbf{s}_t \in \mathbb{R}^n$ (in LocPRB, $\text{supp}(\mathbf{s}_t) \subseteq \mathcal{V}_t$), $\mathbf{f}_t^{(c)}$ is the propagated score vector. PPR is recovered by choosing the geometric weights $c_k = (1 - \alpha)\alpha^k$ (up to the standard convention of whether α denotes continuation or restart probability), which yields $\mathbf{K}_{t-1}^{\text{PPR}} = (1 - \alpha)(\mathbf{I} - \alpha \mathbf{P}_{t-1})^{-1}$.

Heat-kernel diffusion. The heat kernel (HK) corresponds to a Poisson mixture of walk lengths with temperature $\tau > 0$: $c_k^{\text{HK}}(\tau) = e^{-\tau} \tau^k / k!$, $k \geq 0$. The resulting diffusion vector is

$$\mathbf{f}_t^{\text{HK}}(\tau) = e^{-\tau} \sum_{k=0}^{\infty} \frac{\tau^k}{k!} \mathbf{P}_{t-1}^k \mathbf{s}_t = \exp(-\tau(\mathbf{I} - \mathbf{P}_{t-1})) \mathbf{s}_t, \quad (22)$$

which is equivalently the solution at time τ of the graph heat equation (random-walk form)

$$\frac{d}{d\tau} \mathbf{f}_t^{\text{HK}}(\tau) = -(\mathbf{I} - \mathbf{P}_{t-1}) \mathbf{f}_t^{\text{HK}}(\tau), \mathbf{f}_t^{\text{HK}}(0) = \mathbf{s}_t. \quad (23)$$

Compared to PPR, HK typically yields a sharper, more localized diffusion profile for appropriate τ and can be preferable when link formation is dominated by short-range topology (Chung, 2007; Kloster & Gleich, 2014; Bai et al., 2024).

5. Experiments

We begin by benchmarking LocPRB against bandit-based and graph-based baselines in *online* link prediction. We then compare the computational efficiency of LocPRB

Dataset	#Nodes	#Edges	Context dim. (d)
MovieLens	12,000	508,572	20
ogbl-Collab	235,868	1,285,465	128
ogbl-PPA	576,289	30,326,273	58
ogbl-Vessel	3,538,495	5,345,897	3

Table 2. Graph dataset statistics used in online link prediction.

and PRB, highlighting the speedups enabled by warm-started DYN-APPR propagation on large graphs. Finally, we present ablations that isolate the contributions of contextual scoring and fast propagation.

5.1. Online Regret Performance

Datasets and Setups. We study online link prediction on two groups of real-world graphs: (i) a recommendation dataset, MovieLens (Harper & Konstan, 2015); and (ii) large-scale OGB link prediction graphs, ogbl-collab, ogbl-ppa, and ogbl-vessel (Hu et al., 2020). For MovieLens, the node set is bipartite with users U and items I (i.e., $\mathcal{V} = U \cup I$); at each round we draw a user $v_t \in U$ and set $\mathcal{V}_t \subseteq I$, where positives are items purchased by v_t . For OGB graphs, $\mathcal{V}$ is unipartite; we draw $v_t \in \mathcal{V}$ and set $\mathcal{V}_t \subseteq \mathcal{V}$, where positives are true neighbors of v_t in the original graph (with their edges hidden until selected). For reference on graph scale, we report dataset statistics in Table 2. We start from a cold graph $\mathcal{G}_0 = (\mathcal{V}, \mathcal{E}_0 = \emptyset)$ and run for $T = 10,000$ rounds on MovieLens and $T = 5,000$ rounds on the OGB datasets. At each round, we sample a candidate pool $\mathcal{V}_t$ of size 10 with 1 ground-truth positives and 9 uniformly sampled negatives. LocPRB selects $\hat{v}_t \in \mathcal{V}_t$, observes binary feedback, and inserts $(v_t, \hat{v}_t)$ iff the reward is 1 (otherwise $\mathcal{G}_t = \mathcal{G}_{t-1}$).

Baselines. We primarily compare against PRB (Ban et al., 2024) since PRB already reports improvements over standard neural contextual bandits such as NEURALGREEDY, NEURALUCB, NEURALTS, and EE-NET (Ban et al., 2022;

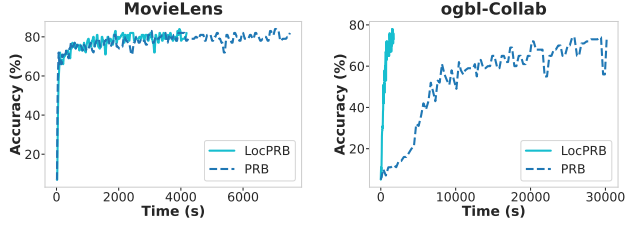


Figure 2. Test accuracy versus time for PRB and LOCPRB on MovieLens and ogbl-Collab, evaluated periodically at fixed checkpoints during online learning (average of 10 runs).

Zhou et al., 2020; Zhang et al., 2021). Following Ban et al. (2024), we use the same network update schedule (every 50 rounds for the first 2,000 rounds and then every 100 rounds thereafter); additional details are deferred to Appendix E.

Results. Figure 1 reports the regret trajectory over rounds, and Table 1 summarizes the final cumulative regret (lower is better). Across all datasets, LOCPRB matches PRB in regret and achieves lower (or comparable) regret on the larger OGB graphs, indicating that local propagation preserves the decision quality of full PageRank-based scoring in online link prediction.

5.2. Speedup over PRB

Figure 2 plots test accuracy as a function of wall-clock time, where accuracy is evaluated periodically at fixed checkpoints during online learning; for end-of-run summaries, see Table 1. Compared with reporting accuracy versus rounds, this view directly captures the practical efficiency of an online method: a curve that rises faster indicates a better accuracy–time tradeoff. Across two datasets, LOCPRB exhibits a consistently more favorable tradeoff than PRB: it attains comparable and often slightly higher test accuracy while requiring less runtime to reach a given performance level. In particular, LOCPRB typically matches the accuracy of PRB at much earlier times and converges to the high-accuracy regime sooner, reflecting that warm-started local pushes reduce the per-round propagation cost without changing the decision rule qualitatively. The speedup is modest on MovieLens, where propagation is not the dominant bottleneck, but becomes pronounced on the large OGB graphs, where full PageRank propagation is substantially more expensive. Overall, the accuracy–time curves confirm that dynamic local propagation delivers substantial practical gains in responsiveness and scalability while preserving predictive performance.

5.3. Ablation Studies

We study how the APPR tolerance ϵ and the PageRank damping factor α affect the accuracy–efficiency tradeoff of LOCPRB on ogbl-Collab. Table 3 varies the propagation tolerance and reports the final cumulative regret. Following common practice in local diffusion, we parameterize the tolerance as $\epsilon = c/n$, where n is the number of nodes and c

$\epsilon (\times 1/n)$	Regret	Time
PRB	2199 ± 114.5	29952 ± 483.3
10000	4568 ± 34.9	2781 ± 11.9
1000	2504 ± 152.8	2956 ± 15.3
100	2271 ± 83.2	2961 ± 22.9
10	2195 ± 105.3	2983 ± 22.5
1	2195 ± 129.5	1676 ± 115.5
10^{-1}	2195 ± 109.4	2832 ± 183.7

Table 3. Effect of the APPR tolerance ϵ on ogbl-Collab: final cumulative regret (mean $\pm$ std over 10 runs). Here “ $\epsilon (\times 1/n)$ ” means the actual tolerance is $(\text{entry})/n$, where n is the number of nodes.

α	PRB		LOCPRB	
	Regret	Time	Regret	Time
0.6	1826 ± 95.5	30731 ± 221.5	1815 ± 89.2	1145 ± 19.9
0.7	1917 ± 99.0	30751 ± 171.9	1920 ± 97.3	1161 ± 18.2
0.8	2068 ± 144.0	30739 ± 222.7	2067 ± 148.7	1580 ± 7.63
0.9	2375 ± 119.7	30278 ± 167.7	2355 ± 105.6	1620 ± 12.8

Table 4. Sensitivity to the PageRank damping factor α on ogbl-Collab: final cumulative regret (mean $\pm$ std over 10 runs) for PRB and LOCPRB.

is a tunable constant; this normalization makes the stopping rule comparable across graphs of different sizes. Our regret analysis uses $\epsilon = \mathcal{O}(1/\sqrt{T})$ to ensure that propagation error does not dominate the learning term. In our experiments, setting $\epsilon = c/n$ with moderate c empirically provides sufficiently accurate propagation while keeping the local-push work small.

Table 3 shows that overly loose tolerances (e.g., $\epsilon = 10000/n$) substantially hurt performance, while moderate tolerances quickly recover the same regret as PRB; in particular, $\epsilon \leq 100/n$ yields essentially identical regret to the full-propagation baseline on ogbl-Collab. Table 4 indicates that the damping factor controls a clear tradeoff: smaller α leads to more localized diffusion and lower regret in this setting (best around $\alpha \in \{0.6, 0.7\}$), whereas larger α over-smooths the propagated scores and degrades regret (e.g., $\alpha = 0.9$).

6. Conclusions

We formulated dynamic link prediction on evolving graphs as a contextual bandit problem and proposed LOCPRB, which combines neural exploitation–exploration scoring with warm-started local diffusion (DYN-APPR) to enable sublinear-time propagation. We proved regret and dynamic-work guarantees that separate learning and propagation errors, and experiments show that LOCPRB improves regret over strong neural bandit baselines while matching PRB’s regret with substantially lower runtime on large graphs. An exciting direction is to generalize the same bandit–propagation coupling to broader diffusion operators and more general graph update regimes.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Notations

Table 5 summarizes the main symbols used throughout the paper. Unless stated otherwise, vectors are column vectors.

Table 5. Summary of Notations (constants, graphs, bandits, and neural networks).

Symbol	Description
Constants and indices	
T	Number of rounds / iterations.
$[T]$	Set of rounds, $[T] = \{1, 2, \dots, T\}$.
k	Candidate-set size per round.
$\mathbf{x}(u)$	Entry of $\mathbf{x}$ indexed by node u .
$\text{supp}(\mathbf{x})$	Support set of $\mathbf{x}$.
Graphs and random walks	
$\alpha \in (0, 1)$	Damping factor in PPR.
n	Number of nodes, $n = \mathcal{V} $.
$\mathcal{G}_0 = (\mathcal{V}, \mathcal{E}_0)$	Initial undirected graph.
$\{\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)\}_{t=0}^{T-1}$	Evolving graph sequence over T rounds.
$\mathcal{N}_t(v)$	Neighborhood of v at time t .
$d_v^{(t)}$	Degree of v at time t .
$\mathbf{A}_t \in \mathbb{R}^{n \times n}$	Adjacency matrix of $\mathcal{G}_t$.
$\mathbf{D}_t \in \mathbb{R}^{n \times n}$	Degree matrix of $\mathcal{G}_t$.
$\mathbf{P}_t \in \mathbb{R}^{n \times n}$	Random-walk transition matrix, $\mathbf{P}_t = \mathbf{D}_t^{-1} \mathbf{A}_t$.
$\mathbf{M}_t \in \mathbb{R}^{n \times n}$	PPR linear-system matrix.
$\mathbf{s}_t \in \mathbb{R}^n$	Personalization vector.
$\boldsymbol{\pi}_t \in \mathbb{R}^n$	Exact PPR vector on $\mathcal{G}_{t-1}$.
$\hat{\boldsymbol{\pi}}_t \in \mathbb{R}^n$	ϵ -approximate PPR vector on $\mathcal{G}_{t-1}$ computed by APPR.
$\boldsymbol{\pi}_t^* \in \mathbb{R}^n$	Oracle PPR vector on $\mathcal{G}_{t-1}$ computed from $\mathbf{y}_t$.
$\mathcal{P}_t(\epsilon, \alpha, \mathbf{s}_t, \mathcal{G}_{t-1})$	Set of ϵ -approximate PPR solutions (cf. (4)).
$\mathbf{p}, \mathbf{r} \in \mathbb{R}^n$	APPR estimate vector and residual vector, respectively.
$\mathbf{e}_u$	Indicator vector of node u .
ξ	Residual mass at an active node in a PUSH, $\xi = \mathbf{r}(u)$.
work_t	Total processed degree over all executed PUSH operations at update t .
$\text{vol}_t(\mathcal{S})$	Volume of $\mathcal{S}$ at time t , $\sum_{v \in \mathcal{S}} d_v^{(t)}$.
Bandits and rewards	
d	Context (feature) dimension, $\mathbf{x}_{t,i} \in \mathbb{R}^d$.
$v_t \in \mathcal{V}$	Serving (source) node at round t .
$\mathcal{V}_t = \{v_{t,1}, \dots, v_{t,k}\}$	Candidate node set at round t , $ \mathcal{V}_t = k$.
$\mathcal{X}_t = \{\mathbf{x}_{t,1}, \dots, \mathbf{x}_{t,k}\}$	Context vectors of candidates in $\mathcal{V}_t$.
$i \in [k]$	Candidate index (arm) at round t .
i_t^*	Optimal arm in $\mathcal{V}_t$, $i_t^* = \arg \max_{i \in [k]} \mathbb{P}(r_{t,i} = 1 \mid \mathbf{x}_{t,i})$.
$\hat{i}_t$	Arm selected by the learner at round t .
$v_{t,\hat{i}_t}$	Selected candidate node at round t .
$r_{t,i} \in \{0, 1\}$	Binary reward for arm i at round t .
$\mathcal{Y} = \{0, 1\}$	Reward space.
$\mathcal{D}_{\mathcal{Y} \mathbf{x}_{t,i}}$	Conditional reward distribution given context $\mathbf{x}_{t,i}$.

Continued on next page

Table 5 (continued)

Symbol	Description
$\mathbf{R}_T$	Cumulative pseudo-regret over T rounds.
Neural networks and learning quantities	
L	Depth (number of layers) of a fully-connected network.
m	Width of hidden layers.
p	Parameter dimension (after vectorization).
$f_1(\cdot; \theta^1)$	Exploitation network.
$f_2(\cdot; \theta^2)$	Exploration network.
$\phi(x)$	Gradient feature.
$\mathbf{W}_\ell$	Weight matrix of layer ℓ .
θ	Vectorized network parameters.
$\mathbf{h}_\ell$	Hidden-layer activations.
$\mathbf{g}_\ell$	Pre-activations.
$\mathbf{D}_\ell$	ReLU mask.
$\mathcal{B}(\theta_0, R)$	Parameter ball.
$\Psi(\theta_0, R)$	Local empirical risk infimum.
LOCPRB	
ϵ	Target accuracy for approximate PPR computation.
η_1, η_2	SGD step sizes for updating θ^1 and θ^2 .
$f_1(\cdot; \theta^1)$	Exploitation network mapping $\mathbf{x}_{t,i}$ to $\mu_{t,i}$.
$f_2(\cdot; \theta^2)$	Exploration network mapping $\phi(\mathbf{x}_{t,i})$ to $b_{t,i}$.
θ_t^1, θ_t^2	Network parameters after updates at round t .
$\mu_{t,i}$	Exploitation score, $\mu_{t,i} = f_1(\mathbf{x}_{t,i}; \theta_{t-1}^1)$.
$\phi(\mathbf{x}_{t,i})$	Gradient feature, $\phi(\mathbf{x}_{t,i}) = \nabla_{\theta^1} f_1(\mathbf{x}_{t,i}; \theta_{t-1}^1)$.
$b_{t,i}$	Exploration bonus, $b_{t,i} = f_2(\phi(\mathbf{x}_{t,i}); \theta_{t-1}^2)$.
$\boldsymbol{\mu}_t \in \mathbb{R}^n$	Sparse node-indexed exploitation vector supported on $\mathcal{V}_t$.
$\mathbf{b}_t \in \mathbb{R}^n$	Sparse node-indexed exploration-bonus vector supported on $\mathcal{V}_t$.
$\mathcal{L}_1(\theta_{t-1}^1)$	Squared loss for f_1 , $\frac{1}{2}(\mu_{t,\hat{i}} - r_{t,\hat{i}})^2$.
$\mathcal{L}_2(\theta_{t-1}^2)$	Squared loss for f_2 , $\frac{1}{2}(b_{t,\hat{i}} - (r_{t,\hat{i}} - \mu_{t,\hat{i}}))^2$.
DYN-APPR	Dynamic APPR update routine for single-edge insertions.
INSERTUPDATE	Constant-time correction routine restoring the invariant after insertion.
INSERTONEDIR	One-direction correction step used inside INSERTUPDATE.
Δ	Local correction magnitude in INSERTONEDIR, $\Delta = \frac{1}{\alpha} \cdot \frac{\mathbf{p}(u)}{d_u}$.
a, b	Endpoints of the inserted undirected edge at an update.
Regret analysis	
$\mathbf{y}(\cdot)$	Unknown reward function, $\mathbb{E}[r_{t,i} \mid \mathbf{x}_{t,i}] = \mathbf{y}(\mathbf{x}_{t,i})$.
$\mathbf{y}_t \in \mathbb{R}^n$	Sparse expected-reward vector supported on $\mathcal{V}_t$.
$\mathbf{y} \in \mathbb{R}^{Tk}$	Stacked reward vector, $\mathbf{y} = [\mathbf{y}(\mathbf{x}_{t,i})]_{t \in [T], i \in [k]}$.
$\mathbf{H} \in \mathbb{R}^{Tk \times Tk}$	NTK matrix on $\{\mathbf{x}_{t,i}\}_{t \in [T], i \in [k]}$.
λ_0	NTK non-degeneracy constant in Assumption 4.3.
ζ	Log-determinant complexity term, $\zeta = \frac{\log \det(\mathbf{I} + \mathbf{H})}{\log(1 + Tk)}$.
S	NTK-induced reward complexity, $S = \sqrt{\mathbf{y}^\top \mathbf{H}^{-1} \mathbf{y}}$.
Basic operations	

Continued on next page

Table 5 (continued)

Symbol	Description
$\ \cdot\ _2$	Euclidean norm.
$\ \cdot\ _F$	Frobenius norm.
$\ \cdot\ _{\infty, D_{t-1}}$	Degree-normalized ℓ_∞ norm.
$\sigma(\cdot)$	ReLU activation.
$\mathbb{1}\{\cdot\}$	Indicator function.

B. Proofs for Section 4.2

The next lemma collects basic operator properties of P_{t-1} and M_{t-1} , and shows how controlling the linear-system residual implies membership in $\mathcal{P}_t(\epsilon, \alpha, s_t, \mathcal{G}_{t-1})$.

Lemma B.1 (Properties of the PPR operator). *Let M_{t-1} be defined as in (3), and let $\pi_t = (1 - \alpha)M_{t-1}^{-1}s_t$. Then the following properties hold.*

- **(Non-expansiveness of P_{t-1})** For any $x \in \mathbb{R}^n$,

$$\|P_{t-1}x\|_{\infty, D_{t-1}} \leq \|x\|_{\infty, D_{t-1}}. \quad (24)$$

- **(Resolvent bound)** For any $x \in \mathbb{R}^n$,

$$\|M_{t-1}^{-1}x\|_{\infty, D_{t-1}} \leq \frac{1}{1 - \alpha} \|x\|_{\infty, D_{t-1}}. \quad (25)$$

- **(Residual-to-error guarantee)** For any $\hat{\pi} \in \mathbb{R}^n$, if

$$\|M_{t-1}\hat{\pi} - (1 - \alpha)s_t\|_{\infty, D_{t-1}} \leq (1 - \alpha)\epsilon,$$

then

$$\|D_{t-1}^{-1}(\hat{\pi} - \pi_t)\|_\infty \leq \epsilon,$$

i.e., $\hat{\pi} \in \mathcal{P}_t(\epsilon, \alpha, s_t, \mathcal{G}_{t-1})$.

Proof of Lemma B.1. Recall that $P_{t-1} \triangleq D_{t-1}^{-1}A_{t-1}$ and $M_{t-1} \triangleq I - (1 - \alpha)P_{t-1}$, where $\alpha \in (0, 1)$. For brevity, throughout this appendix we drop the subscript $(t - 1)$ and write A, D, P, M . For a vector $x \in \mathbb{R}^n$, recall the degree-normalized ℓ_∞ norm

$$\|x\|_{\infty, D} \triangleq \max_{u \in \mathcal{V}} \frac{|x(u)|}{d_u},$$

where d_u denotes the (current) degree of node u and $D = \text{diag}(d_1, \dots, d_n)$. (When isolated nodes exist, the statements below apply verbatim on the support $\{u : d_u > 0\}$; in our setting we assume $d_u > 0$ for all u .) We prove each item in turn.

(i) Non-expansiveness of P . By definition, $P = D^{-1}A$ is a row-stochastic matrix. For any $x \in \mathbb{R}^n$ and any node $u \in \mathcal{V}$,

$$\begin{aligned} \frac{|(Px)(u)|}{d_u} &= \frac{1}{d_u} \left| \sum_{v \in \mathcal{V}} \frac{A_{uv}}{d_u} x(v) \right| \leq \frac{1}{d_u} \sum_{v \in \mathcal{V}} \frac{A_{uv}}{d_u} |x(v)| \\ &\leq \left(\max_{v \in \mathcal{V}} \frac{|x(v)|}{d_v} \right) \sum_{v \in \mathcal{V}} \frac{A_{uv}}{d_u} = \|x\|_{\infty, D}, \end{aligned}$$

where we used $A_{uv} \geq 0$ and $\sum_v A_{uv} = d_u$. Taking the maximum over u yields $\|Px\|_{\infty, D} \leq \|x\|_{\infty, D}$.

(ii) Resolvent bound. Consider the induced operator norm $\|B\|_{\infty, D} \triangleq \sup_{x \neq 0} \|Bx\|_{\infty, D} / \|x\|_{\infty, D}$. By part (i), $\|P\|_{\infty, D} \leq 1$, hence

$$\|(1 - \alpha)P\|_{\infty, D} \leq 1 - \alpha < 1.$$

Therefore the Neumann series converges and

$$\mathbf{M}^{-1} = (\mathbf{I} - (1 - \alpha)\mathbf{P})^{-1} = \sum_{k=0}^{\infty} (1 - \alpha)^k \mathbf{P}^k. \quad (26)$$

Using sub-multiplicativity of induced norms and again $\|\mathbf{P}\|_{\infty, D} \leq 1$, for any $\mathbf{x} \in \mathbb{R}^n$ we have

$$\|\mathbf{M}^{-1}\mathbf{x}\|_{\infty, D} \leq \sum_{k=0}^{\infty} (1 - \alpha)^k \|\mathbf{P}^k \mathbf{x}\|_{\infty, D} \leq \sum_{k=0}^{\infty} (1 - \alpha)^k \|\mathbf{x}\|_{\infty, D} = \frac{1}{\alpha} \|\mathbf{x}\|_{\infty, D},$$

which is the desired resolvent bound.

(iii) Residual-to-error guarantee. Let π_t be the exact PPR vector, i.e., $\mathbf{M}\pi_t = \alpha \mathbf{s}_t$, and let $\hat{\pi}$ be any approximation. Define the residual $\mathbf{r} \triangleq \mathbf{M}\hat{\pi} - \alpha \mathbf{s}_t$ and the error $\mathbf{e} \triangleq \hat{\pi} - \pi_t$. By subtracting $\mathbf{M}\pi_t = \alpha \mathbf{s}_t$ from $\mathbf{M}\hat{\pi} = \alpha \mathbf{s}_t + \mathbf{r}$ we get

$$\mathbf{M}\mathbf{e} = \mathbf{r} \implies \mathbf{e} = \mathbf{M}^{-1}\mathbf{r}.$$

Taking the degree-normalized ℓ_{∞} norm and applying part (ii),

$$\|\mathbf{e}\|_{\infty, D} = \|\mathbf{M}^{-1}\mathbf{r}\|_{\infty, D} \leq \frac{1}{\alpha} \|\mathbf{r}\|_{\infty, D}.$$

Hence if $\|\mathbf{r}\|_{\infty, D} \leq \alpha\epsilon$, then $\|\mathbf{e}\|_{\infty, D} \leq \epsilon$. Finally, note that

$$\|\mathbf{D}^{-1}\mathbf{e}\|_{\infty} = \max_{u \in \mathcal{V}} \frac{|e(u)|}{d_u} = \|\mathbf{e}\|_{\infty, D},$$

so $\|\mathbf{D}^{-1}(\hat{\pi} - \pi_t)\|_{\infty} \leq \epsilon$, i.e., $\hat{\pi} \in \mathcal{P}_t(\epsilon, \alpha, \mathbf{s}_t, \mathcal{G}_{t-1})$ as defined in (4). This completes the proof. $\square$

Lemma B.2 (ℓ_1 stability of the random-walk and PPR operators). *Let $\mathbf{P}_{t-1} = \mathbf{D}_{t-1}^{-1}\mathbf{A}_{t-1}$ and $\mathbf{M}_{t-1} = \mathbf{I} - \alpha\mathbf{P}_{t-1}$. Then:*

- (ℓ_1 non-expansiveness of $\mathbf{P}_{t-1}$) For any $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{P}_{t-1}\mathbf{x}\|_1 \leq \|\mathbf{x}\|_1. \quad (27)$$

- (ℓ_1 bound on the PPR solution) Let $\pi_t = (1 - \alpha)\mathbf{M}_{t-1}^{-1}\mathbf{s}_t$ be the PPR vector defined by (2). Then

$$\|\pi_t\|_1 \leq \|\mathbf{s}_t\|_1. \quad (28)$$

Proof. For (27), note that $\mathbf{P}_{t-1}$ is row-stochastic, i.e., $\sum_v (\mathbf{P}_{t-1})_{uv} = 1$ and $(\mathbf{P}_{t-1})_{uv} \geq 0$. Hence

$$\|\mathbf{P}_{t-1}\mathbf{x}\|_1 = \sum_u \left| \sum_v (\mathbf{P}_{t-1})_{uv} x(v) \right| \leq \sum_u \sum_v (\mathbf{P}_{t-1})_{uv} |x(v)| = \sum_v |x(v)| \sum_u (\mathbf{P}_{t-1})_{uv} \leq \sum_v |x(v)| = \|\mathbf{x}\|_1,$$

where the last inequality uses $\sum_u (\mathbf{P}_{t-1})_{uv} \leq 1$ for $\mathbf{P}_{t-1} = \mathbf{D}_{t-1}^{-1}\mathbf{A}_{t-1}$ on an undirected graph. For (28), $\pi_t = \alpha\mathbf{P}_{t-1}\pi_t + (1 - \alpha)\mathbf{s}_t$ yields

$$\|\pi_t\|_1 \leq \alpha\|\mathbf{P}_{t-1}\pi_t\|_1 + (1 - \alpha)\|\mathbf{s}_t\|_1 \leq \alpha\|\pi_t\|_1 + (1 - \alpha)\|\mathbf{s}_t\|_1,$$

and hence $\|\pi_t\|_1 \leq \|\mathbf{s}_t\|_1$. $\square$

Lemma B.3 (Linear-system invariant of PUSH). *Let $\mathbf{P}_{t-1}$ be the random-walk matrix on $\mathcal{G}_{t-1}$ and $\mathbf{M}_{t-1} \triangleq \mathbf{I} - \alpha\mathbf{P}_{t-1}$ for some $\alpha \in (0, 1)$. Assume $(\mathbf{p}, \mathbf{r})$ satisfies the invariant*

$$\mathbf{M}_{t-1}\mathbf{p} + (1 - \alpha)\mathbf{r} = (1 - \alpha)\mathbf{s}_t.$$

Let $(\mathbf{p}^+, \mathbf{r}^+)$ be obtained from applying one PUSH at node u :

$$\mathbf{p}^+ = \mathbf{p} + (1 - \alpha)\mathbf{r}(u)\mathbf{e}_u, \quad \mathbf{r}^+ = \mathbf{r} - \mathbf{r}(u)\mathbf{e}_u + \alpha\mathbf{r}(u)\mathbf{P}_{t-1}\mathbf{e}_u.$$

Then $(\mathbf{p}^+, \mathbf{r}^+)$ also satisfies the same invariant:

$$\mathbf{M}_{t-1}\mathbf{p}^+ + (1 - \alpha)\mathbf{r}^+ = (1 - \alpha)\mathbf{s}_t.$$

Proof of Lemma B.3. Let $\xi \triangleq \mathbf{r}(u)$. By the PUSH update,

$$\mathbf{p}^+ - \mathbf{p} = (1 - \alpha) \xi \mathbf{e}_u, \quad \mathbf{r}^+ - \mathbf{r} = -\xi \mathbf{e}_u + \alpha \xi \mathbf{P}_{t-1} \mathbf{e}_u.$$

Consider the change in the left-hand side of the invariant:

$$\begin{aligned} \mathbf{M}_{t-1} \mathbf{p}^+ + (1 - \alpha) \mathbf{r}^+ - (\mathbf{M}_{t-1} \mathbf{p} + (1 - \alpha) \mathbf{r}) &= \mathbf{M}_{t-1} (\mathbf{p}^+ - \mathbf{p}) + (1 - \alpha) (\mathbf{r}^+ - \mathbf{r}) \\ &= \mathbf{M}_{t-1} ((1 - \alpha) \xi \mathbf{e}_u) + (1 - \alpha) (-\xi \mathbf{e}_u + \alpha \xi \mathbf{P}_{t-1} \mathbf{e}_u) \\ &= (1 - \alpha) \xi (\mathbf{M}_{t-1} \mathbf{e}_u - \mathbf{e}_u + \alpha \mathbf{P}_{t-1} \mathbf{e}_u). \end{aligned}$$

Using $\mathbf{M}_{t-1} = \mathbf{I} - \alpha \mathbf{P}_{t-1}$, we have

$$\mathbf{M}_{t-1} \mathbf{e}_u - \mathbf{e}_u + \alpha \mathbf{P}_{t-1} \mathbf{e}_u = (\mathbf{I} - \alpha \mathbf{P}_{t-1}) \mathbf{e}_u - \mathbf{e}_u + \alpha \mathbf{P}_{t-1} \mathbf{e}_u = \mathbf{0}.$$

Therefore the change is zero, i.e.,

$$\mathbf{M}_{t-1} \mathbf{p}^+ + (1 - \alpha) \mathbf{r}^+ = \mathbf{M}_{t-1} \mathbf{p} + (1 - \alpha) \mathbf{r}.$$

If $(\mathbf{p}, \mathbf{r})$ satisfies the invariant with right-hand side $(1 - \alpha) \mathbf{s}_t$, then so does $(\mathbf{p}^+, \mathbf{r}^+)$. This completes the proof. $\square$

Theorem B.4 (Restatement of Theorem 4.1). *Run APPR (Algorithm 2) on $\mathcal{G}_{t-1}$ with threshold ϵ and activity rule $|\mathbf{r}(u)| \geq \epsilon d_u^{(t-1)}$. Then:*

- **(Termination)** *The algorithm terminates after finitely many PUSH operations.*
- **(Accuracy)** *Upon termination, $\|\mathbf{r}\|_{\infty, D_{t-1}} \leq \epsilon$ and the returned estimate $\hat{\pi}_t := \mathbf{p}$ satisfies (4).*
- **(Locality)** *The total processed degree and explored volume satisfy*

$$\text{work}_t \text{ and } \text{vol}_{t-1}(\text{supp}(\hat{\pi}_t)) \leq \frac{1}{(1 - \alpha)\epsilon}.$$

Proof of Theorem 4.1. Fix round t and abbreviate $\mathbf{P} := \mathbf{P}_{t-1}$, $\mathbf{D} := \mathbf{D}_{t-1}$, and $\mathbf{M} := \mathbf{M}_{t-1} = \mathbf{I} - \alpha \mathbf{P}$ with $\alpha \in (0, 1)$. Let π_t be the exact PPR vector, i.e., the unique solution to

$$\pi_t = \alpha \mathbf{P} \pi_t + (1 - \alpha) \mathbf{s}_t, \quad \text{equivalently} \quad \mathbf{M} \pi_t = (1 - \alpha) \mathbf{s}_t.$$

We prove termination, accuracy, and locality for APPR.

Consider one PUSH at node u with $\xi := \mathbf{r}(u)$. The update (12) implies

$$\mathbf{r}^+ = \mathbf{r} - \xi \mathbf{e}_u + \alpha \xi \mathbf{P} \mathbf{e}_u, \quad \text{and} \quad \mathbf{p}^+ = \mathbf{p} + (1 - \alpha) \xi \mathbf{e}_u.$$

Since $\mathbf{P} \mathbf{e}_u$ is supported on $\mathcal{N}_{t-1}(u)$ and has nonnegative entries summing to 1, we have $\|\mathbf{P} \mathbf{e}_u\|_1 = 1$. By the triangle inequality,

$$\begin{aligned} \|\mathbf{r}^+\|_1 &= \|\mathbf{r} - \xi \mathbf{e}_u + \alpha \xi \mathbf{P} \mathbf{e}_u\|_1 \\ &\leq \|\mathbf{r} - \xi \mathbf{e}_u\|_1 + \alpha |\xi| \|\mathbf{P} \mathbf{e}_u\|_1 \\ &= (\|\mathbf{r}\|_1 - |\xi|) + \alpha |\xi| = \|\mathbf{r}\|_1 - (1 - \alpha) |\xi|. \end{aligned} \tag{29}$$

Therefore, every PUSH decreases $\|\mathbf{r}\|_1$ by at least $(1 - \alpha) |\mathbf{r}(u)|$.

A node u is pushed only if it is active, i.e., $|\mathbf{r}(u)| \geq \epsilon d_u^{(t-1)}$. Combining with (29) gives

$$\|\mathbf{r}^+\|_1 \leq \|\mathbf{r}\|_1 - (1 - \alpha) \epsilon d_u^{(t-1)}. \tag{30}$$

Let work_t be the total processed degree across all PUSH operations, i.e., $\text{work}_t = \sum d_u^{(t-1)}$ over every pushed node u . Summing (30) over all pushes and using $\|\mathbf{r}\|_1 \geq 0$ yields

$$(1 - \alpha) \epsilon \text{work}_t \leq \|\mathbf{r}^{(0)}\|_1 - \|\mathbf{r}^{(\text{final})}\|_1 \leq \|\mathbf{r}^{(0)}\|_1.$$

Under a cold start, $\mathbf{r}^{(0)} = \mathbf{s}_t$, and by our normalization $\|\mathbf{s}_t\|_1 = 1$. Hence

$$\text{work}_t \leq \frac{1}{(1-\alpha)\epsilon}. \quad (31)$$

In particular, the algorithm can execute only finitely many PUSH operations, which proves **Termination**. Moreover, when the loop stops, no active node remains, i.e., $\|\mathbf{r}\|_{\infty, D} \leq \epsilon$.

Let $(\mathbf{p}, \mathbf{r})$ be the pair at termination and set $\hat{\boldsymbol{\pi}}_t := \mathbf{p}$. The PUSH update preserves the linear-system invariant (13):

$$M\mathbf{p} + (1-\alpha)\mathbf{r} = (1-\alpha)\mathbf{s}_t.$$

Rearranging gives

$$M\hat{\boldsymbol{\pi}}_t - (1-\alpha)\mathbf{s}_t = -(1-\alpha)\mathbf{r},$$

and therefore

$$\|M\hat{\boldsymbol{\pi}}_t - (1-\alpha)\mathbf{s}_t\|_{\infty, D} = (1-\alpha)\|\mathbf{r}\|_{\infty, D} \leq (1-\alpha)\epsilon.$$

Applying the residual-to-error guarantee in Lemma B.1 yields

$$\|D^{-1}(\hat{\boldsymbol{\pi}}_t - \boldsymbol{\pi}_t)\|_{\infty} \leq \epsilon,$$

which is exactly $\hat{\boldsymbol{\pi}}_t \in \mathcal{P}_t(\epsilon, \alpha, \mathbf{s}_t, \mathcal{G}_{t-1})$ in (4). This proves **Accuracy**.

Each PUSH at node u can only create nonzero mass on u and its neighbors, so after the algorithm terminates, every node in $\text{supp}(\hat{\boldsymbol{\pi}}_t)$ must have been either pushed or received residual from some pushed node. Hence the explored node set is contained in the union of neighborhoods of pushed nodes, and its volume is bounded by the total processed degree:

$$\text{vol}_{t-1}(\text{supp}(\hat{\boldsymbol{\pi}}_t)) \leq \sum_{u \text{ pushed}} d_u^{(t-1)} = \text{work}_t.$$

Combining with (31) yields the **Locality** statement:

$$\text{vol}_{t-1}(\text{supp}(\hat{\boldsymbol{\pi}}_t)) \leq \text{work}_t \leq \frac{1}{(1-\alpha)\epsilon}.$$

□

Theorem B.5 (Restatement of Theorem 4.2). *Consider a graph sequence $\{\mathcal{G}_t\}_{t=1}^T$ where each $\mathcal{G}_t$ is obtained from $\mathcal{G}_{t-1}$ by inserting at most one undirected edge. For each update t , run DYN-APPR (Algorithm 3) with $\mathbf{s}_t$. Let $(\mathbf{p}, \mathbf{r})$ denote the maintained estimate-residual state after the INSERTUPDATE step and before the subsequent APPR call, and let work_t be the total processed degree of all PUSH operations executed by that APPR call.*

- **(Invariant preservation)** The INSERTUPDATE step produces a state $(\mathbf{p}, \mathbf{r})$ satisfying the linear-system invariant (13)
- **(Total push work bound)** If each inserted edge has a uniformly random endpoint in $\mathcal{V}$, then

$$\mathbb{E}\left[\sum_{t=1}^T \text{work}_t\right] = \mathcal{O}\left(\frac{1}{(1-\alpha)\epsilon} + \frac{T}{(1-\alpha)^2} + \frac{T}{n(1-\alpha)^2\epsilon}\right).$$

In particular, for constant α and $\epsilon = \Omega(1/n)$,

$$\mathbb{E}\left[\sum_{t=1}^T \text{work}_t\right] = \mathcal{O}\left(\frac{T}{(1-\alpha)^2} + \frac{1}{(1-\alpha)\epsilon}\right).$$

Proof of Theorem 4.2. For each $t \in [T]$, let $(\mathbf{p}_t, \mathbf{r}_t)$ denote the estimate-residual pair after the final APPR call in update t (i.e., after DYN-APPR finishes update t). Let $(\mathbf{p}_t^{(0)}, \mathbf{r}_t^{(0)})$ be the pair immediately after INSERTUPDATE and before the subsequent APPR call in update t , and let work_t be the total processed degree across all PUSH operations executed by that APPR call.

We show that one call to INSERTONEDIR preserves the linear-system invariant (13). Fix a graph $\mathcal{G}$ and let $\mathcal{G}^+$ be obtained by inserting a single *directed* edge (u, v) (i.e., only the outgoing neighborhood of u changes). Write $\mathbf{P}$ and $\mathbf{P}^+$ for the corresponding random-walk operators, and $\mathbf{M} := \mathbf{I} - \alpha\mathbf{P}$, $\mathbf{M}^+ := \mathbf{I} - \alpha\mathbf{P}^+$. Let d_u^- and d_u be the degrees of u before/after insertion, so $d_u = d_u^- + 1$. Since only the transition rule out of u changes, for the basis vector $\mathbf{e}_u$ we have the exact identity

$$\mathbf{P}^+ \mathbf{e}_u = \frac{d_u^-}{d_u} \mathbf{P} \mathbf{e}_u + \frac{1}{d_u} \mathbf{e}_v, \quad \text{and} \quad \mathbf{P}^+ \mathbf{e}_w = \mathbf{P} \mathbf{e}_w \quad (\forall w \neq u). \quad (32)$$

Assume the pre-update pair $(\mathbf{p}, \mathbf{r})$ satisfies the invariant on $\mathcal{G}$,

$$\mathbf{M} \mathbf{p} + (1 - \alpha) \mathbf{r} = (1 - \alpha) \mathbf{s}. \quad (33)$$

INSERTONEDIR first rescales

$$\mathbf{p}^+ = \mathbf{p} + \left(\frac{d_u}{d_u^-} - 1 \right) \mathbf{p}(u) \mathbf{e}_u = \mathbf{p} + \frac{\mathbf{p}(u)}{d_u^-} \mathbf{e}_u,$$

so that $\mathbf{p}^+(u) = \mathbf{p}(u) \cdot \frac{d_u}{d_u^-}$ and $\mathbf{p}^+(w) = \mathbf{p}(w)$ for $w \neq u$. Using (32) and the fact that only the u -coordinate of $\mathbf{p}$ changes, we can expand $\mathbf{P}^+ \mathbf{p}^+$ as

$$\begin{aligned} \mathbf{P}^+ \mathbf{p}^+ &= \mathbf{P}^+ \mathbf{p} + \frac{\mathbf{p}(u)}{d_u^-} \mathbf{P}^+ \mathbf{e}_u \\ &= \mathbf{P} \mathbf{p} + \mathbf{p}(u) (\mathbf{P}^+ \mathbf{e}_u - \mathbf{P} \mathbf{e}_u) + \frac{\mathbf{p}(u)}{d_u^-} \left(\frac{d_u^-}{d_u} \mathbf{P} \mathbf{e}_u + \frac{1}{d_u} \mathbf{e}_v \right) \\ &= \mathbf{P} \mathbf{p} + \mathbf{p}(u) \left(-\frac{1}{d_u} \mathbf{P} \mathbf{e}_u + \frac{1}{d_u} \mathbf{e}_v \right) + \frac{\mathbf{p}(u)}{d_u} \mathbf{P} \mathbf{e}_u + \frac{\mathbf{p}(u)}{d_u d_u^-} \mathbf{e}_v \\ &= \mathbf{P} \mathbf{p} + \frac{\mathbf{p}(u)}{d_u^-} \cdot \frac{1}{d_u} \mathbf{e}_v. \end{aligned}$$

Consequently,

$$\begin{aligned} \mathbf{M}^+ \mathbf{p}^+ - \mathbf{M} \mathbf{p} &= (\mathbf{p}^+ - \alpha \mathbf{P}^+ \mathbf{p}^+) - (\mathbf{p} - \alpha \mathbf{P} \mathbf{p}) \\ &= \frac{\mathbf{p}(u)}{d_u^-} \mathbf{e}_u - \alpha \cdot \frac{\mathbf{p}(u)}{d_u^-} \cdot \frac{1}{d_u} \mathbf{e}_v. \end{aligned} \quad (34)$$

Now INSERTONEDIR updates the residual by

$$\Delta := \frac{1}{\alpha} \cdot \frac{\mathbf{p}^+(u)}{d_u} = \frac{1}{\alpha} \cdot \frac{1}{d_u} \cdot \mathbf{p}(u) \frac{d_u}{d_u^-} = \frac{1}{\alpha} \cdot \frac{\mathbf{p}(u)}{d_u^-},$$

and sets

$$\mathbf{r}^+ = \mathbf{r} - \Delta \mathbf{e}_u + (1 - \alpha) \Delta \mathbf{e}_v.$$

Multiplying by $(1 - \alpha)$ and substituting the above Δ gives

$$(1 - \alpha)(\mathbf{r}^+ - \mathbf{r}) = -(1 - \alpha) \frac{\mathbf{p}(u)}{\alpha d_u^-} \mathbf{e}_u + (1 - \alpha)^2 \frac{\mathbf{p}(u)}{\alpha d_u^-} \mathbf{e}_v. \quad (35)$$

Finally, combining (33), (34), and (35), we obtain

$$\begin{aligned} \mathbf{M}^+ \mathbf{p}^+ + (1 - \alpha) \mathbf{r}^+ - (1 - \alpha) \mathbf{s} &= (\mathbf{M} \mathbf{p} + (1 - \alpha) \mathbf{r} - (1 - \alpha) \mathbf{s}) + (\mathbf{M}^+ \mathbf{p}^+ - \mathbf{M} \mathbf{p}) + (1 - \alpha)(\mathbf{r}^+ - \mathbf{r}) \\ &= \mathbf{0} + \left(\frac{\mathbf{p}(u)}{d_u^-} \mathbf{e}_u - \alpha \cdot \frac{\mathbf{p}(u)}{d_u^-} \cdot \frac{1}{d_u} \mathbf{e}_v \right) + \left(-(1 - \alpha) \frac{\mathbf{p}(u)}{\alpha d_u^-} \mathbf{e}_u + (1 - \alpha)^2 \frac{\mathbf{p}(u)}{\alpha d_u^-} \mathbf{e}_v \right) \\ &= \mathbf{0}, \end{aligned}$$

where the last equality holds by the definition of Δ (equivalently, by direct coefficient matching on $\mathbf{e}_u$ and $\mathbf{e}_v$). Therefore, one INSERTONEDIR step preserves (13) on $\mathcal{G}^+$. Since INSERTUPDATE applies INSERTONEDIR twice (for (a, b) and (b, a)), the invariant is preserved on the undirected insertion as claimed.

We next bound the total push work. Consider one PUSH applied to an active node u on $\mathcal{G}_{t-1}$ with $\xi := \mathbf{r}(u)$. As in (12), the residual update satisfies

$$\mathbf{r}^+ = \mathbf{r} - \xi \mathbf{e}_u + \alpha \xi \mathbf{P} \mathbf{e}_u.$$

Since $\mathbf{P} \mathbf{e}_u \geq 0$ entrywise and $\|\mathbf{P} \mathbf{e}_u\|_1 = 1$, the triangle inequality yields

$$\|\mathbf{r}^+\|_1 \leq \|\mathbf{r}\|_1 - (1 - \alpha)|\xi|.$$

If u is active then $|\xi| \geq \epsilon d_u^{(t-1)}$, hence each push at u decreases $\|\mathbf{r}\|_1$ by at least $(1 - \alpha)\epsilon d_u^{(t-1)}$. Summing over all pushed nodes in update t gives

$$\text{work}_t = \sum_{u \text{ pushed at } t} d_u^{(t-1)} \leq \frac{\|\mathbf{r}_t^{(0)}\|_1 - \|\mathbf{r}_t\|_1}{(1 - \alpha)\epsilon} \leq \frac{\|\mathbf{r}_t^{(0)}\|_1}{(1 - \alpha)\epsilon}. \quad (36)$$

Let the inserted undirected edge at update t be (a_t, b_t) , so $\mathcal{G}_{t-1}$ is obtained from $\mathcal{G}_{t-2}$ by inserting (a_t, b_t) . Since INSERTUPDATE only modifies coordinates at a_t and b_t , we have

$$\|\mathbf{r}_t^{(0)}\|_1 \leq \|\mathbf{r}_{t-1}\|_1 + \|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1. \quad (37)$$

For a single directed correction INSERTONEDIR(u, v), only $\mathbf{r}(u)$ and $\mathbf{r}(v)$ are changed. Writing $\Delta = \frac{1}{\alpha} \frac{\mathbf{p}(u)}{d_u}$ as in Algorithm 3, the update adds $-\Delta$ to $\mathbf{r}(u)$ and adds $(1 - \alpha)\Delta$ to $\mathbf{r}(v)$. Therefore, for an absolute constant $c_0 > 0$ (absorbing the rescaling factor $\frac{d_u}{d_u - 1}$ and $1/\alpha$),

$$\|\Delta \mathbf{r}\|_1 \leq c_0 \cdot \frac{\mathbf{p}(u)}{(1 - \alpha) d_u^{(t-1)}}. \quad (38)$$

Applying (38) to both directions in INSERTUPDATE yields, for another absolute constant $c_1 > 0$,

$$\|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1 \leq \frac{c_1}{1 - \alpha} \left(\frac{\mathbf{p}_{t-1}(a_t)}{d_{a_t}^{(t-1)}} + \frac{\mathbf{p}_{t-1}(b_t)}{d_{b_t}^{(t-1)}} \right). \quad (39)$$

Let π_{t-1} denote the exact PPR solution on $\mathcal{G}_{t-2}$ with injection $\mathbf{s}_{t-1}$. Since $\mathbf{p}_{t-1}$ is an ϵ -approximate PPR vector in the degree-normalized ℓ_∞ sense, we have for all $u \in \mathcal{V}$,

$$\frac{\mathbf{p}_{t-1}(u)}{d_u^{(t-2)}} \leq \frac{\pi_{t-1}(u)}{d_u^{(t-2)}} + \epsilon. \quad (40)$$

Let U_t be a uniformly random endpoint of (a_t, b_t) . By symmetry, $\mathbb{E}[f(a_t) + f(b_t)] = 2\mathbb{E}[f(U_t)]$ for any f , and thus

$$\mathbb{E} \left[\frac{\mathbf{p}_{t-1}(a_t)}{d_{a_t}^{(t-1)}} + \frac{\mathbf{p}_{t-1}(b_t)}{d_{b_t}^{(t-1)}} \right] \leq 2\mathbb{E}_{U \sim \text{Unif}(\mathcal{V})} \left[\frac{\pi_{t-1}(U)}{d_U^{(t-1)}} \right] + 2\epsilon. \quad (41)$$

Using Lemma B.2 and $\|\mathbf{s}_{t-1}\|_1 = 1$, we have $\|\pi_{t-1}\|_1 \leq 1$, and since $d_u^{(t-1)} \geq 1$,

$$\mathbb{E}_{U \sim \text{Unif}(\mathcal{V})} \left[\frac{\pi_{t-1}(U)}{d_U^{(t-1)}} \right] = \frac{1}{n} \sum_{u \in \mathcal{V}} \frac{\pi_{t-1}(u)}{d_u^{(t-1)}} \leq \frac{1}{n} \sum_{u \in \mathcal{V}} \pi_{t-1}(u) = \frac{\|\pi_{t-1}\|_1}{n} \leq \frac{1}{n}. \quad (42)$$

Combining (39)–(42) gives

$$\mathbb{E} \left[\|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1 \right] \leq \mathcal{O} \left(\frac{1}{1 - \alpha} \left(\frac{1}{n} + \epsilon \right) \right). \quad (43)$$

Finally, we sum the per-update work bounds. Plugging (37) into (36) and summing over $t = 1, \dots, T$ yields

$$\begin{aligned} \sum_{t=1}^T \text{work}_t &\leq \frac{1}{(1 - \alpha)\epsilon} \sum_{t=1}^T \|\mathbf{r}_t^{(0)}\|_1 \\ &\leq \frac{1}{(1 - \alpha)\epsilon} \sum_{t=1}^T \left(\|\mathbf{r}_{t-1}\|_1 + \|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1 \right) \\ &= \frac{\|\mathbf{r}_0\|_1}{(1 - \alpha)\epsilon} + \frac{1}{(1 - \alpha)\epsilon} \sum_{t=1}^T \|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1, \end{aligned} \quad (44)$$

where the $\|\mathbf{r}_{t-1}\|_1$ terms telescope using $\|\mathbf{r}_T\|_1 \geq 0$. For $t = 1$, DYN-APPR is a cold start, so $\mathbf{r}_0 = \mathbf{s}_1$ and $\|\mathbf{r}_0\|_1 = \|\mathbf{s}_1\|_1 = 1$. Taking expectations in (44) and applying (43) yields

$$\mathbb{E} \left[\sum_{t=1}^T \text{work}_t \right] \leq \mathcal{O} \left(\frac{1}{(1-\alpha)\epsilon} + \frac{T}{(1-\alpha)^2} \cdot \frac{1}{\epsilon} \left(\frac{1}{n} + \epsilon \right) \right) = \mathcal{O} \left(\frac{1}{(1-\alpha)\epsilon} + \frac{T}{(1-\alpha)^2} + \frac{T}{n(1-\alpha)^2\epsilon} \right),$$

which is the claimed bound. The “in particular” statement follows immediately when α is constant and $\epsilon = \Omega(1/n)$. $\square$

Proof of Theorem 4.2. For each $t \in [T]$, let $(\mathbf{p}_t, \mathbf{r}_t)$ denote the pair after the final APPR call in update t (i.e., after DYN-APPR finishes update t). Let $\mathbf{r}_t^{(0)}$ be the residual vector at the beginning of that final APPR call (i.e., immediately after INSERTUPDATE), and let work_t be the total processed degree across all PUSH operations executed by this APPR call.

Consider one PUSH applied to an active node u on $\mathcal{G}_{t-1}$ with $\xi := \mathbf{r}(u)$. Exactly as in (12), the residual update satisfies

$$\mathbf{r}^+ = \mathbf{r} - \xi \mathbf{e}_u + \alpha \xi \mathbf{P}_{t-1} \mathbf{e}_u.$$

Since $\|\mathbf{P}_{t-1} \mathbf{e}_u\|_1 = 1$ and $\mathbf{P}_{t-1} \mathbf{e}_u \geq 0$ entrywise, the triangle inequality gives

$$\|\mathbf{r}^+\|_1 \leq \|\mathbf{r}\|_1 - (1-\alpha)|\xi|.$$

If u is active then $|\xi| \geq \epsilon d_u^{(t-1)}$, hence each push at u decreases $\|\mathbf{r}\|_1$ by at least $(1-\alpha)\epsilon d_u^{(t-1)}$. Summing over all pushed nodes in update t yields

$$\text{work}_t = \sum_{u \text{ pushed at } t} d_u^{(t-1)} \leq \frac{\|\mathbf{r}_t^{(0)}\|_1 - \|\mathbf{r}_t\|_1}{(1-\alpha)\epsilon} \leq \frac{\|\mathbf{r}_t^{(0)}\|_1}{(1-\alpha)\epsilon}. \quad (45)$$

Let the inserted undirected edge at update t be (a_t, b_t) , so $\mathcal{G}_{t-1}$ is obtained from $\mathcal{G}_{t-2}$ by inserting (a_t, b_t) . DYN-APPR warm-starts from $(\mathbf{p}_{t-1}, \mathbf{r}_{t-1})$ and then applies INSERTUPDATE, producing $(\mathbf{p}_t^{(0)}, \mathbf{r}_t^{(0)})$.

Since INSERTUPDATE consists of two calls to INSERTONEDIR, it changes only coordinates associated with a_t and b_t , and therefore

$$\|\mathbf{r}_t^{(0)}\|_1 \leq \|\mathbf{r}_{t-1}\|_1 + \|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1. \quad (46)$$

For a single directed correction INSERTONEDIR(u, v), define the residual increment vector

$$\Delta \mathbf{r} \triangleq \mathbf{r}^+ - \mathbf{r},$$

where $\mathbf{r}$ and $\mathbf{r}^+$ denote the residual before/after this INSERTONEDIR call. Only $\mathbf{r}(u)$ and $\mathbf{r}(v)$ are modified, and a direct inspection of Algorithm 3 shows that

$$|\Delta \mathbf{r}(u)| \leq \frac{c}{\alpha} \cdot \frac{\mathbf{p}(u)}{d_u^{(t-1)}}, \quad |\Delta \mathbf{r}(v)| \leq c \cdot \frac{\mathbf{p}(u)}{d_u^{(t-1)}},$$

for an absolute constant $c > 0$ (absorbing the factor $\frac{d_u}{d_u-1}$). Hence there is an absolute constant c_0 such that

$$\|\Delta \mathbf{r}\|_1 \leq c_0 \cdot \frac{\mathbf{p}(u)}{(1-\alpha) d_u^{(t-1)}}. \quad (47)$$

Applying (47) to both directions in INSERTUPDATE yields

$$\|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1 \leq \frac{c_1}{1-\alpha} \left(\frac{\mathbf{p}_{t-1}(a_t)}{d_{a_t}^{(t-1)}} + \frac{\mathbf{p}_{t-1}(b_t)}{d_{b_t}^{(t-1)}} \right), \quad (48)$$

for another absolute constant c_1 .

Let π_{t-1} denote the exact PPR solution on $\mathcal{G}_{t-2}$ with injection $\mathbf{s}_{t-1}$. Since $\mathbf{p}_{t-1}$ is an ϵ -approximate PPR vector in the degree-normalized ℓ_∞ sense,

$$\frac{\mathbf{p}_{t-1}(u)}{d_u^{(t-2)}} \leq \frac{\pi_{t-1}(u)}{d_u^{(t-2)}} + \epsilon \quad (\forall u \in \mathcal{V}). \quad (49)$$

Let U_t be a uniformly random endpoint of (a_t, b_t) . Then $\mathbb{E}[f(a_t) + f(b_t)] = 2\mathbb{E}[f(U_t)]$ for any f , and thus

$$\mathbb{E}\left[\frac{\mathbf{p}_{t-1}(a_t)}{d_{a_t}^{(t-1)}} + \frac{\mathbf{p}_{t-1}(b_t)}{d_{b_t}^{(t-1)}}\right] \leq 2\mathbb{E}_{U \sim \text{Unif}(\mathcal{V})}\left[\frac{\pi_{t-1}(U)}{d_U^{(t-1)}}\right] + 2\epsilon. \quad (50)$$

Next, using Lemma B.2 (ℓ_1 bound on the PPR solution), $\|\pi_{t-1}\|_1 \leq \|\mathbf{s}_{t-1}\|_1 = 1$ under our normalization. Since $d_u^{(t-1)} \geq 1$,

$$\mathbb{E}_{U \sim \text{Unif}(\mathcal{V})}\left[\frac{\pi_{t-1}(U)}{d_U^{(t-1)}}\right] = \frac{1}{n} \sum_{u \in \mathcal{V}} \frac{\pi_{t-1}(u)}{d_u^{(t-1)}} \leq \frac{1}{n} \sum_{u \in \mathcal{V}} \pi_{t-1}(u) = \frac{\|\pi_{t-1}\|_1}{n} \leq \frac{1}{n}. \quad (51)$$

Combining (48)–(51) gives

$$\mathbb{E}\left[\|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1\right] \leq \mathcal{O}\left(\frac{1}{1-\alpha} \left(\frac{1}{n} + \epsilon\right)\right). \quad (52)$$

Plugging (46) into (45) and summing over $t = 1, \dots, T$ yields

$$\begin{aligned} \sum_{t=1}^T \text{work}_t &\leq \frac{1}{(1-\alpha)\epsilon} \sum_{t=1}^T \|\mathbf{r}_t^{(0)}\|_1 \\ &\leq \frac{1}{(1-\alpha)\epsilon} \sum_{t=1}^T \left(\|\mathbf{r}_{t-1}\|_1 + \|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1\right) \\ &= \frac{\|\mathbf{r}_0\|_1}{(1-\alpha)\epsilon} + \frac{1}{(1-\alpha)\epsilon} \sum_{t=1}^T \|\mathbf{r}_t^{(0)} - \mathbf{r}_{t-1}\|_1, \end{aligned} \quad (53)$$

where the $\|\mathbf{r}_{t-1}\|_1$ terms telescope using $\|\mathbf{r}_T\|_1 \geq 0$. For $t = 1$, DYN-APPR is a cold start, so $\mathbf{r}_0 = \mathbf{s}_1$ and $\|\mathbf{r}_0\|_1 = \|\mathbf{s}_1\|_1 = 1$.

Taking expectations in (53) and applying (52) gives

$$\mathbb{E}\left[\sum_{t=1}^T \text{work}_t\right] \leq \mathcal{O}\left(\frac{1}{(1-\alpha)\epsilon} + \frac{T}{(1-\alpha)^2} \cdot \frac{1}{\epsilon} \left(\frac{1}{n} + \epsilon\right)\right) = \mathcal{O}\left(\frac{1}{(1-\alpha)\epsilon} + \frac{T}{(1-\alpha)^2} + \frac{T}{n(1-\alpha)^2\epsilon}\right),$$

which is the claimed bound. The “in particular” statement follows immediately when α is constant and $\epsilon = \Omega(1/n)$. $\square$

C. Neural Tangent Kernel

Theorem 4.4 is associated with the Neural Tangent Kernel (NTK) matrix as follows: given the context set $\{\mathbf{x}_{t,i}\}_{t \in [T], i \in [k]}$, we define the induced depth- L NTK matrix $\mathbf{H}$. For completeness, we record below the standard NTK definition and the non-degeneracy assumption used in neural contextual bandit analyses.

Definition C.1 (Neural Tangent Kernel (NTK) (Jacot et al., 2018; Arora et al., 2019)). Let $\mathcal{N}$ denote the normal distribution. Given the context set $\{\mathbf{x}_{t,i}\}_{t \in [T], i \in [k]}$, we flatten the double index (t, i) into a single index $s \in [Tk]$, and write the resulting list as $\{\mathbf{x}_s\}_{s=1}^{Tk}$. Define recursively for any $s, s' \in [Tk]$:

$$\begin{aligned} \Sigma_{s,s'}^0 &= \langle \mathbf{x}_s, \mathbf{x}_{s'} \rangle, \\ \mathbf{A}_{s,s'}^{\ell-1} &= \begin{pmatrix} \Sigma_{s,s}^{\ell-1} & \Sigma_{s,s'}^{\ell-1} \\ \Sigma_{s',s}^{\ell-1} & \Sigma_{s',s'}^{\ell-1} \end{pmatrix}, \\ \Sigma_{s,s'}^\ell &= 2\mathbb{E}_{(a,b) \sim \mathcal{N}(0, \mathbf{A}_{s,s'}^{\ell-1})}[\sigma(a)\sigma(b)], \end{aligned} \quad (54)$$

$$\mathbf{H}_{s,s'}^\ell = 2\mathbf{H}_{s,s'}^{\ell-1} \mathbb{E}_{(a,b) \sim \mathcal{N}(0, \mathbf{A}_{s,s'}^{\ell-1})}[\sigma'(a)\sigma'(b)] + \Sigma_{s,s'}^\ell, \quad (55)$$

with initialization $\mathbf{H}_{s,s'}^0 = \Sigma_{s,s'}^0$. The induced NTK matrix is then defined as

$$\mathbf{H} = \frac{\mathbf{H}^L + \Sigma^L}{2} \in \mathbb{R}^{Tk \times Tk}. \quad (56)$$

Assumption C.2 (NTK non-degeneracy). There exists $\lambda_0 > 0$ such that $\mathbf{H} \succeq \lambda_0 \mathbf{I}$.

D. Proofs for Theorem 4.4

D.1. Preliminaries

We follow the standard over-parameterized NTK analysis used in neural contextual bandits. Consider a fully-connected ReLU network $f(\cdot; \theta)$ with depth L and width m . Given an input $\mathbf{x}$, define the hidden-layer activations as

$$\mathbf{h}_0 = \mathbf{x}, \quad \mathbf{h}_\ell = \sigma(\mathbf{W}_\ell \mathbf{h}_{\ell-1}), \ell \in [L-1].$$

We further define the binary diagonal gating matrices induced by ReLU:

$$\mathbf{D}_\ell = \text{diag}(\mathbb{1}\{(\mathbf{W}_\ell \mathbf{h}_{\ell-1})_1 > 0\}, \dots, \mathbb{1}\{(\mathbf{W}_\ell \mathbf{h}_{\ell-1})_m > 0\}), \ell \in [L-1].$$

Accordingly, the network output admits the matrix product form

$$f(\mathbf{x}; \theta) = \mathbf{W}_L \left(\prod_{\ell=1}^{L-1} \mathbf{D}_\ell \mathbf{W}_\ell \right) \mathbf{x}. \quad (57)$$

Throughout the paper, both the exploitation network f_1 and the exploration network f_2 are instantiated from the same architectural form (57), with independent parameters and (possibly) different inputs. Moreover, we assume that m , η_1 , and η_2 satisfy the conditions in Theorem 4.4.

We define the (parameter) gradient of the network output by

$$\mathbf{g}(\mathbf{x}; \theta) \triangleq \nabla_\theta f(\mathbf{x}; \theta) \in \mathbb{R}^p. \quad (58)$$

In particular, the layer-wise gradients admit the following closed forms:

$$\nabla_{\mathbf{W}_\ell} f(\mathbf{x}; \theta) = \begin{cases} \left[\mathbf{h}_{\ell-1} \mathbf{W}_L \left(\prod_{r=\ell+1}^{L-1} \mathbf{D}_r \mathbf{W}_r \right) \mathbf{D}_\ell \right]^\top, & \ell \in [L-1], \\ \mathbf{h}_{L-1}^\top, & \ell = L. \end{cases} \quad (59)$$

Let $\{(\mathbf{x}_{t,i}, r_{t,i})\}_{t \in [T], i \in [k]}$ denote all context–reward pairs across T rounds. For each round $t \in [T]$ and arm $i \in [k]$, define the squared loss

$$\mathcal{L}_{t,i}(\theta) \triangleq \frac{1}{2} (f(\mathbf{x}_{t,i}; \theta) - r_{t,i})^2.$$

When the arm index is unambiguous, we write $\mathcal{L}_t(\theta) \equiv \mathcal{L}_{t,i}(\theta)$ for an arbitrary $i \in [k]$.

Given a radius $R > 0$, define the local parameter neighborhood around initialization θ_0 as

$$\mathcal{B}(\theta_0, R) \triangleq \left\{ \theta : \|\theta - \theta_0\|_2 \leq R/m^{1/4} \right\}. \quad (60)$$

We define the instance-dependent approximation error

$$\Psi(\theta_0, R) \triangleq \inf_{\theta \in \mathcal{B}(\theta_0, R)} \sum_{t=1}^T \sum_{i=1}^k (f(\mathbf{x}_{t,i}; \theta) - r_{t,i})^2 \equiv \inf_{\theta \in \mathcal{B}(\theta_0, R)} \sum_{t=1}^T \sum_{i=1}^k 2\mathcal{L}_{t,i}(\theta). \quad (61)$$

For each round $t \in [T]$ and arm $i \in [k]$, define exploitation and exploration predictions as

$$\mu_{t,i} \triangleq f_1(\mathbf{x}_{t,i}; \theta_{t-1}^1), \quad b_{t,i} \triangleq f_2(\phi(\mathbf{x}_{t,i}); \theta_{t-1}^2), \quad (62)$$

so that the realized prediction error is $\mu_{t,i} + b_{t,i} - r_{t,i}$.

The following auxiliary lemmas will be used in our analysis.

Lemma D.1 (Uniform bounds in a local neighborhood). *With probability at least $1 - \mathcal{O}(TkL) \exp(-\Omega(m\omega^{2/3}L))$ over the random initialization, the following event holds simultaneously for all $(t, i) \in [T] \times [k]$ and all θ satisfying $\|\theta - \theta_0\|_2 \leq \omega$, where $\omega \leq \mathcal{O}(L^{-9/2}(\log m)^{-3})$:*

(a) **Output bound:** $|f(\mathbf{x}; \theta)| \leq \mathcal{O}(1)$.

(b) **Gradient bound:** $\|g(\mathbf{x}; \theta)\|_2 = \|\nabla_\theta f(\mathbf{x}; \theta)\|_2 \leq \mathcal{O}(\sqrt{L})$.

(c) **Loss-gradient bound:** $\|\nabla_\theta \mathcal{L}_t(\theta)\|_2 \leq \mathcal{O}(\sqrt{L})$.

Lemma D.2 (Second-order remainder of linearization). *Suppose the conditions in Lemma D.1 hold. With probability at least $1 - \mathcal{O}(TkL) \exp(-\Omega(m\omega^{2/3}L))$ over the random initialization, for all $(t, i) \in [T] \times [k]$ and all $\theta, \theta' \in \mathcal{B}(\theta_0, \omega m^{1/4})$ with $\omega \leq \mathcal{O}(L^{-9/2}(\log m)^{-3})$, it holds uniformly that*

$$\left| f(\mathbf{x}; \theta) - f(\mathbf{x}; \theta') - \langle \nabla_{\theta'} f(\mathbf{x}; \theta'), \theta - \theta' \rangle \right| \leq \mathcal{O}(\omega^{1/3} L^2 \sqrt{\log m}) \|\theta - \theta'\|_2. \quad (63)$$

Lemma D.3 (Local Lipschitzness). *With probability at least $1 - \mathcal{O}(TkL) \exp(-\Omega(m\omega^{2/3}L))$ over the random initialization, for all $\theta, \theta' \in \mathcal{B}(\theta_0, \omega m^{1/4})$ with $\omega \leq \mathcal{O}(L^{-9/2}(\log m)^{-3})$, it holds uniformly that*

$$|f(\mathbf{x}; \theta) - f(\mathbf{x}; \theta')| \leq \mathcal{O}(\omega \sqrt{L}) + \mathcal{O}(\omega^{4/3} L^2 \sqrt{\log m}). \quad (64)$$

Lemma D.4 (Almost convexity of the squared loss). *With probability at least $1 - \mathcal{O}(TkL^2) \exp(-\Omega(m\omega^{2/3}L))$ over the random initialization, for all $t \in T$ and all θ, θ' satisfying $\|\theta - \theta_0\|_2 \leq \omega$ and $\|\theta' - \theta_0\|_2 \leq \omega$ with $\omega \leq \mathcal{O}(L^{-6}(\log m)^{-3/2})$, it holds uniformly that*

$$\mathcal{L}_t(\theta') \geq \mathcal{L}_t(\theta) + \langle \nabla \mathcal{L}_t(\theta), \theta' - \theta \rangle - \varepsilon_{ac}, \quad (65)$$

where $\varepsilon_{ac} = \mathcal{O}(\omega^{4/3} L^3 \sqrt{\log m})$.

Lemma D.5 (Parameter trajectory stays in the local ball). *With probability at least $1 - \mathcal{O}(TkL^2) \exp[-\Omega(m\omega^{2/3}L)]$ over the randomness of θ_0 , for any $R > 0$, it holds uniformly that*

$$\|\theta_t - \theta_0\|_2 \leq \mathcal{B}(\theta_0, R) \triangleq \left\{ \theta : \|\theta - \theta_0\|_2 \leq R/m^{1/4} \right\}, \quad \forall t \in [T]. \quad (66)$$

Lemma D.6 (Instance-dependent loss bound). *Define $\theta^* \in \arg \inf_{\theta \in \mathcal{B}(\theta_0, R)} \sum_{t=1}^T \mathcal{L}_t(\theta)$. With probability at least $1 - \mathcal{O}(TkL^2) \exp[-\Omega(m\omega^{2/3}L)]$ over the randomness of θ_0 , given any $R > 0$, it holds that*

$$\sum_{t=1}^T \mathcal{L}_t(\theta_t) \leq \sum_{t=1}^T \mathcal{L}_t(\theta^*) + \mathcal{O}(1) + \mathcal{O}\left(\frac{TLR^2}{\sqrt{m}}\right) + \mathcal{O}\left(\frac{TR^{4/3} L^2 \sqrt{\log m}}{m^{1/3}}\right). \quad (67)$$

D.2. A key martingale bound for the exploration network

Lemma D.7 (Online prediction error bound). *Let $\hat{i} \in [k]$ be the arm selected by the Algorithm 1 at round t , and let $\mathcal{H}_t = \{\mathbf{x}_{\tau, \hat{i}}, r_{\tau, \hat{i}}\}_{\tau=1}^t$ denote the history up to round t . Then for any $\delta \in (0, 1)$ and any $R > 0$, with probability at least $1 - \delta$, it holds simultaneously for all $t \in [T]$ that*

$$\frac{1}{t} \sum_{\tau=1}^t \mathbb{E} \left[|\mu_{\tau, \hat{i}} + b_{\tau, \hat{i}} - r_{\tau, \hat{i}}| \mid \mathcal{H}_{\tau-1} \right] \leq \sqrt{\frac{\Psi(\theta_0, R) + \mathcal{O}(1)}{t}} + \sqrt{\frac{2 \log(1/\delta)}{t}}, \quad (68)$$

Proof. Fix any $R > 0$ and $t \in [T]$. By Lemma D.5, with probability at least $1 - \delta$, the iterates satisfy $\theta_0^2, \dots, \theta_{t-1}^2 \in \mathcal{B}(\theta_0, R)$. Moreover, by Lemma D.1, the predictions are uniformly bounded, i.e., for all $\tau \in [t]$,

$$|\mu_{\tau, \hat{i}} + b_{\tau, \hat{i}} - r_{\tau, \hat{i}}| \leq \mathcal{O}(1). \quad (69)$$

For any $\tau \in [t]$, define the martingale difference

$$V_\tau := \mathbb{E}_{r_{\tau, \hat{i}}} \left[|b_{\tau, \hat{i}} - (r_{\tau, \hat{i}} - \mu_{\tau, \hat{i}})| \right] - |b_{\tau, \hat{i}} - (r_{\tau, \hat{i}} - \mu_{\tau, \hat{i}})|. \quad (70)$$

By construction and the definition of the history $\mathcal{H}_{\tau-1}$, we have

$$\mathbb{E}[V_\tau \mid \mathcal{H}_{\tau-1}] = 0,$$

and thus $\{V_\tau\}_{\tau=1}^t$ is a martingale difference sequence. Using (69) and Hoeffding–Azuma inequality, with probability at least $1 - \delta$,

$$\frac{1}{t} \sum_{\tau=1}^t \mathbb{E}_{r_{\tau,i}} [|b_{\tau,i} - (r_{\tau,i} - \mu_{\tau,i})|] \leq \frac{1}{t} \sum_{\tau=1}^t |b_{\tau,i} - (r_{\tau,i} - \mu_{\tau,i})| + \sqrt{\frac{2 \log(1/\delta)}{t}}. \quad (71)$$

Next we bound the empirical average on the right-hand side of (71). By Cauchy–Schwarz,

$$\frac{1}{t} \sum_{\tau=1}^t |b_{\tau,i} - (r_{\tau,i} - \mu_{\tau,i})| \leq \frac{1}{\sqrt{t}} \left(\sum_{\tau=1}^t (b_{\tau,i} - (r_{\tau,i} - \mu_{\tau,i}))^2 \right)^{1/2}. \quad (72)$$

Applying Lemma D.6, for any $\theta' \in \mathcal{B}(\theta_0, R)$, with probability at least $1 - \delta$,

$$\begin{aligned} \sum_{\tau=1}^t \mathcal{L}_{t,i}(\theta^1, \theta^2) &\leq \sum_{\tau=1}^t \left(f_2(\phi(\mathbf{x}_{\tau,i}); \theta') - (r_{\tau,i} - \mu_{\tau,i}) \right)^2 + \mathcal{O}(1) \\ &\leq \Psi(\theta_0, R) + \mathcal{O}(1), \end{aligned} \quad (73)$$

where the last step follows from the definition of $\Psi(\theta_0, R)$. Combining (72) and (73) yields

$$\frac{1}{t} \sum_{\tau=1}^t |b_{\tau,i} - (r_{\tau,i} - \mu_{\tau,i})| \leq \sqrt{\frac{\Psi(\theta_0, R) + \mathcal{O}(1)}{t}}. \quad (74)$$

Finally, plugging (74) into (71) gives that, with probability at least $1 - \delta$,

$$\frac{1}{t} \sum_{\tau=1}^t \mathbb{E} \left[|\mu_{\tau,i} + b_{\tau,i} - r_{\tau,i}| \mid \mathcal{H}_{\tau-1} \right] \leq \sqrt{\frac{\Psi(\theta_0, R) + \mathcal{O}(1)}{t}} + \sqrt{\frac{2 \log(1/\delta)}{t}}.$$

A standard union bound over $t \in [T]$ completes the proof. $\square$

Corollary D.8 (Prediction error bound under a fixed policy). *Fix any policy π^* , and for any $t \in [T]$, let $i^* \in [k]$ denote the arm selected by π^* with corresponding reward r_{τ,i^*} at round τ . Let $\{\theta_{\tau-1}^1, \theta_{\tau-1}^2\}_{\tau=1}^t$ be the intermediate parameters produced by Algorithm 1 when trained on the data generated by π^* . Denote by $\mathcal{H}_{\tau-1}^*$ the history up to round $\tau - 1$ under π^* . Then for any $\delta \in (0, 1)$ and any $R > 0$, with probability at least $1 - \delta$, it holds simultaneously for all $t \in [T]$ that*

$$\frac{1}{t} \sum_{\tau=1}^t \mathbb{E} \left[|\mu_{\tau,i^*} + b_{\tau,i^*} - r_{\tau,i^*}| \mid \pi^*, \mathcal{H}_{\tau-1}^* \right] \leq \sqrt{\frac{\Psi(\theta_0, R) + \mathcal{O}(1)}{t}} + \sqrt{\frac{2 \log(1/\delta)}{t}}. \quad (75)$$

D.3. Ridge/NTK approximation ingredients

Definition D.9 (Design matrices and an auxiliary iterate sequence). Given the selected arm sequence $\{i_t\}_{t=1}^T$, define the matrices and reward vector

$$\begin{aligned} \mathbf{G}^{(0)} &:= [\mathbf{g}(\mathbf{x}_{1,i_1}; \theta_0), \dots, \mathbf{g}(\mathbf{x}_{T,i_T}; \theta_0)] \in \mathbb{R}^{p \times T}, \\ \mathbf{G}_0 &:= [\mathbf{g}(\mathbf{x}_{1,1}; \theta_0), \dots, \mathbf{g}(\mathbf{x}_{T,k}; \theta_0)] \in \mathbb{R}^{p \times Tk}, \\ \mathbf{r} &:= (r_{1,i_1}, \dots, r_{T,i_T})^\top \in \mathbb{R}^T. \end{aligned}$$

Here $\mathbf{G}^{(0)}$ and $\mathbf{r}$ are constructed from the selected context–reward pairs $\{(\mathbf{x}_{t,i_t}, r_{t,i_t})\}_{t=1}^T$, whereas $\mathbf{G}_0$ stacks all gradients $\{\mathbf{g}(\mathbf{x}_{t,i}; \theta_0)\}_{t \in [T], i \in [k]}$ as columns.

With stepsize $\eta = m^{-1/4}$, define the auxiliary sequence $\{\theta^{(j)}\}_{j \geq 0}$ by

$$\theta^{(0)} = \theta_0, \quad \theta^{(j+1)} = \theta^{(j)} - \eta \left[\mathbf{G}^{(0)} \left((\mathbf{G}^{(0)})^\top (\theta^{(j)} - \theta_0) - \mathbf{r} \right) + \lambda (\theta^{(j)} - \theta_0) \right].$$

Lemma D.10 (Concentration of the empirical NTK matrix). *For any $\delta \in (0, 1)$ and $\epsilon > 0$, if*

$$m = \Omega\left(\frac{L^6 \log(TkL/\delta)}{(\epsilon/Tk)^4}\right),$$

then with probability at least $1 - \delta$, the following holds:

$$\|\mathbf{G}_0 \mathbf{G}_0^\top - \mathbf{H}\|_F \leq \epsilon.$$

Proof. Fix $\epsilon > 0$ and $\delta \in (0, 1)$. By Theorem 3.1 of [Arora et al. \(2019\)](#), there exists an absolute constant such that if

$$m = \Omega\left(\frac{L^6 \log(L/\delta)}{\epsilon^4}\right),$$

then with probability at least $1 - \delta$ over the random initialization θ_0 , for any pair $(i, j) \in [Tk] \times [Tk]$,

$$|\langle \mathbf{G}_0^\top \mathbf{G}_0 \rangle_{i,j} - \mathbf{H}_{i,j}| \leq \epsilon. \quad (76)$$

Taking a union bound over all $(i, j) \in [Tk] \times [Tk]$ implies that if

$$m = \Omega\left(\frac{L^6 \log(TkL/\delta)}{\epsilon^4}\right),$$

then with probability at least $1 - \delta$ the above inequality holds simultaneously for all (i, j) .

Under this event, we have

$$\|\mathbf{G}_0^\top \mathbf{G}_0 - \mathbf{H}\|_F^2 = \sum_{i=1}^{Tk} \sum_{j=1}^{Tk} |(\mathbf{G}_0^\top \mathbf{G}_0)_{i,j} - \mathbf{H}_{i,j}|^2 \leq (Tk)^2 \epsilon^2,$$

and hence $\|\mathbf{G}_0^\top \mathbf{G}_0 - \mathbf{H}\|_F \leq Tk \epsilon$.

Finally, setting $\epsilon = \epsilon/(Tk)$ yields $\|\mathbf{G}_0^\top \mathbf{G}_0 - \mathbf{H}\|_F \leq \epsilon$. Substituting this choice into the width requirement gives

$$m = \Omega\left(\frac{L^6 \log(TkL/\delta)}{(\epsilon/(Tk))^4}\right),$$

which completes the proof. $\square$

Lemma D.11 (Representation by NTK features). *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random initialization θ_0 , the following holds. For any $\mathbf{y} \in \mathbb{R}^{Tk}$, there exists $\theta^* \in \mathbb{R}^p$ such that*

$$\mathbf{G}_0^\top (\theta^* - \theta_0) = \mathbf{y}, \quad (77)$$

equivalently, $\langle g(\mathbf{x}_{t,i}; \theta_0), \theta^ - \theta_0 \rangle = \mathbf{y}_{t,i}$ for all $(t, i) \in [T] \times [k]$. Moreover, one may choose θ^* satisfying*

$$\|\theta^* - \theta_0\|_2^2 \leq 2 \mathbf{y}^\top \mathbf{H}^{-1} \mathbf{y} \triangleq 2S^2.$$

Proof. Let $\epsilon = \lambda_0/2$. By Lemma D.10, with probability at least $1 - \delta$,

$$\|\mathbf{G}_0^\top \mathbf{G}_0 - \mathbf{H}\|_F \leq \epsilon.$$

Therefore,

$$\mathbf{G}_0^\top \mathbf{G}_0 \succeq \mathbf{H} - \|\mathbf{G}_0^\top \mathbf{G}_0 - \mathbf{H}\|_F \mathbf{I} \succeq \mathbf{H} - \frac{\lambda_0}{2} \mathbf{I} \succeq \frac{1}{2} \mathbf{H} \succeq 0,$$

and hence $\mathbf{G}_0^\top \mathbf{G}_0$ is positive semidefinite. Moreover, since $\mathbf{H} \succeq \lambda_0 \mathbf{I}$, we also have $\mathbf{G}_0^\top \mathbf{G}_0 \succeq \mathbf{H}/2 \succ 0$, so $(\mathbf{G}_0^\top \mathbf{G}_0)^{-1}$ exists.

Now fix any $\mathbf{y} \in \mathbb{R}^{Tk}$ and consider the singular value decomposition

$$\mathbf{G}_0 = \mathbf{P} \mathbf{A} \mathbf{Q}^\top,$$

where $\mathbf{P} \in \mathbb{R}^{p \times Tk}$ has orthonormal columns, $\mathbf{Q} \in \mathbb{R}^{Tk \times Tk}$ is orthogonal, and $\mathbf{A} \in \mathbb{R}^{Tk \times Tk}$ is diagonal with nonnegative entries. Define

$$\theta^* := \theta_0 + \mathbf{P}\mathbf{A}^{-1}\mathbf{Q}^\top \mathbf{y}.$$

Then

$$\mathbf{G}_0^\top (\theta^* - \theta_0) = \mathbf{Q}\mathbf{A}\mathbf{P}^\top \mathbf{P}\mathbf{A}^{-1}\mathbf{Q}^\top \mathbf{y} = \mathbf{y},$$

which is exactly (77), i.e., $\langle g(\mathbf{x}_{t,i}; \theta_0), \theta^* - \theta_0 \rangle = \mathbf{y}_{t,i}$ for all $(t, i) \in [T] \times [k]$.

Finally,

$$\|\theta^* - \theta_0\|_2^2 = \mathbf{y}^\top \mathbf{Q}\mathbf{A}^{-2}\mathbf{Q}^\top \mathbf{y} = \mathbf{y}^\top (\mathbf{G}_0^\top \mathbf{G}_0)^{-1} \mathbf{y} \leq 2 \mathbf{y}^\top \mathbf{H}^{-1} \mathbf{y},$$

where the inequality uses $\mathbf{G}_0^\top \mathbf{G}_0 \succeq \mathbf{H}/2$. $\square$

Lemma D.12 (Existence of a near-optimal GD iterate). *Let $\{\theta^{(j)}\}_{j \geq 0}$ be the auxiliary sequence defined in Definition D.9 with stepsize $\eta = m^{-1/4}$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random initialization θ_0 , there exist an index $j \geq 0$ and $\theta' := \theta^{(j)}$ such that*

$$(a) \quad \|\theta' - \theta_0\|_2 \leq \frac{\tilde{\mathcal{O}}(T^{3/2}L + \sqrt{T})}{m^{1/4}};$$

$$(b) \quad \|\theta' - \theta_0 - \hat{\theta}_t\|_2 \leq \frac{1}{1 + \mathcal{O}(TL)}.$$

Proof. Recall that $\theta^{(0)} = \theta_0$ and $\eta = m^{-1/4}$. Consider the ridge objective

$$\mathcal{L}(\theta) := \frac{1}{2} \|(\mathbf{G}^{(0)})^\top (\theta - \theta_0) - \mathbf{r}\|_2^2 + \frac{\lambda}{2} \|\theta - \theta_0\|_2^2. \quad (78)$$

A direct calculation gives

$$\nabla \mathcal{L}(\theta) = \mathbf{G}^{(0)} ((\mathbf{G}^{(0)})^\top (\theta - \theta_0) - \mathbf{r}) + \lambda(\theta - \theta_0),$$

so the auxiliary sequence $\{\theta^{(j)}\}_{j \geq 0}$ is exactly gradient descent on $\mathcal{L}$.

We start with part (a). By Lemma D.1, $\|g(\mathbf{x}_{t,i_t}; \theta_0)\|_2 \leq \mathcal{O}(\sqrt{L})$ uniformly, and hence

$$\|\mathbf{G}^{(0)}\|_F^2 = \sum_{t=1}^T \|g(\mathbf{x}_{t,i_t}; \theta_0)\|_2^2 \leq \mathcal{O}(TL), \quad \text{so} \quad \|\mathbf{G}^{(0)}\|_F \leq \mathcal{O}(\sqrt{TL}). \quad (79)$$

Using the update rule and telescoping, for any $j \geq 1$,

$$\|\theta^{(j)} - \theta_0\|_2 \leq \sum_{s=0}^{j-1} \|\theta^{(s+1)} - \theta^{(s)}\|_2 = \eta \sum_{s=0}^{j-1} \|\nabla \mathcal{L}(\theta^{(s)})\|_2. \quad (80)$$

To bound the gradients, rewrite

$$\mathbf{G}^{(0)} ((\mathbf{G}^{(0)})^\top (\theta - \theta_0) - \mathbf{r}) = \mathbf{G}^{(0)} (\mathbf{G}^{(0)})^\top (\theta - \theta_0) - \mathbf{G}^{(0)} \mathbf{r},$$

so that for any $s \geq 0$,

$$\begin{aligned} \|\nabla \mathcal{L}(\theta^{(s)})\|_2 &\leq \|\mathbf{G}^{(0)} (\mathbf{G}^{(0)})^\top (\theta^{(s)} - \theta_0)\|_2 + \|\mathbf{G}^{(0)} \mathbf{r}\|_2 + \lambda \|\theta^{(s)} - \theta_0\|_2 \\ &\leq \|\mathbf{G}^{(0)}\|_2^2 \|\theta^{(s)} - \theta_0\|_2 + \|\mathbf{G}^{(0)}\|_2 \|\mathbf{r}\|_2 + \lambda \|\theta^{(s)} - \theta_0\|_2. \end{aligned} \quad (81)$$

Next we upper bound $\|\theta^{(s)} - \theta_0\|_2$ uniformly in s using the objective value. From (78), $\mathcal{L}(\theta) \geq \frac{\lambda}{2} \|\theta - \theta_0\|_2^2$. Moreover, for $\eta \leq 1/(\|\mathbf{G}^{(0)}\|_2^2 + \lambda)$, gradient descent decreases $\mathcal{L}$, hence $\mathcal{L}(\theta^{(s)}) \leq \mathcal{L}(\theta^{(0)}) = \frac{1}{2} \|\mathbf{r}\|_2^2$ for all s . Combining these gives

$$\|\theta^{(s)} - \theta_0\|_2 \leq \frac{\|\mathbf{r}\|_2}{\sqrt{\lambda}} \quad \forall s \geq 0. \quad (82)$$

We also have $\|\mathbf{G}^{(0)}\|_2^2 \leq \|\mathbf{G}^{(0)}\|_F^2 \leq \mathcal{O}(TL)$ by (79), and $\|\mathbf{r}\|_2 \leq \mathcal{O}(\sqrt{T})$ since rewards are bounded. Substituting these bounds and (82) into (81) yields, for all s ,

$$\|\nabla \mathcal{L}(\theta^{(s)})\|_2 \leq \mathcal{O}(TL) \cdot \mathcal{O}(\sqrt{T/\lambda}) + \mathcal{O}(\sqrt{TL}) \cdot \mathcal{O}(\sqrt{T}) + \mathcal{O}(\sqrt{T\lambda}).$$

Finally, the middle term is dominated by the other two via AM–GM:

$$\mathcal{O}(\sqrt{TL}\sqrt{T}) = \mathcal{O}(T\sqrt{L}) \leq \mathcal{O}(TL\sqrt{T/\lambda} + \sqrt{T\lambda}).$$

Therefore,

$$\|\nabla \mathcal{L}(\theta^{(s)})\|_2 \leq \mathcal{O}(TL\sqrt{T/\lambda} + \sqrt{T\lambda}) \quad \forall s.$$

Plugging this into (80) gives

$$\|\theta^{(j)} - \theta_0\|_2 \leq \eta j \cdot \mathcal{O}(TL\sqrt{T/\lambda} + \sqrt{T\lambda}) = \frac{\mathcal{O}(j(TL\sqrt{T/\lambda} + \sqrt{T\lambda}))}{m^{1/4}}.$$

The objective $\mathcal{L}$ is a λ -strongly convex quadratic with Hessian $(\mathbf{G}^{(0)})(\mathbf{G}^{(0)})^\top + \lambda \mathbf{I}$, and thus has the unique minimizer $\theta^* = \theta_0 + \hat{\theta}_t$. For $\eta \leq 1/(\|\mathbf{G}^{(0)}\|_2^2 + \lambda)$, standard results for gradient descent on λ -strongly convex and smooth functions imply linear convergence of the objective gap:

$$\mathcal{L}(\theta^{(j)}) - \mathcal{L}(\theta^*) \leq (1 - \eta\lambda)^j (\mathcal{L}(\theta^{(0)}) - \mathcal{L}(\theta^*)).$$

By λ -strong convexity,

$$\|\theta^{(j)} - \theta^*\|_2^2 \leq \frac{2}{\lambda} (\mathcal{L}(\theta^{(j)}) - \mathcal{L}(\theta^*)),$$

and therefore

$$\begin{aligned} \|\theta^{(j)} - \theta_0 - \hat{\theta}_t\|_2^2 &= \|\theta^{(j)} - \theta^*\|_2^2 \\ &\leq \frac{2}{\lambda} (1 - \eta\lambda)^j (\mathcal{L}(\theta^{(0)}) - \mathcal{L}(\theta^*)) \leq \frac{2}{\lambda} (1 - \eta\lambda)^j \mathcal{L}(\theta^{(0)}). \end{aligned}$$

Since $\theta^{(0)} = \theta_0$, we have $\mathcal{L}(\theta^{(0)}) = \frac{1}{2} \|\mathbf{r}\|_2^2$, and hence

$$\|\theta^{(j)} - \theta_0 - \hat{\theta}_t\|_2^2 \leq \frac{(1 - \eta\lambda)^j}{\lambda} \|\mathbf{r}\|_2^2 \leq \frac{T(1 - \eta\lambda)^j}{\lambda}, \quad (83)$$

where the last inequality uses bounded rewards, i.e., $\|\mathbf{r}\|_2^2 \leq T$. Finally, setting $\lambda = 1$ and choosing j such that $(1 - \eta)^j \leq (T(1 + \mathcal{O}(TL)))^{-1}$, equivalently, $j \geq \frac{\log(T(1 + \mathcal{O}(TL)))}{-\log(1 - \eta)}$, we obtain from (83) that

$$\|\theta^{(j)} - \theta_0 - \hat{\theta}_t\|_2^2 \leq \frac{1}{1 + \mathcal{O}(TL)}.$$

Taking $\theta' := \theta^{(j)}$ concludes the proof of (b) and completes the proof. $\square$

Definition D.13 (NTK ridge regression estimator). Let $\{i_t\}_{t=1}^T$ denote the selected-arm sequence, and let $\{(\mathbf{x}_{t,i_t}, r_{t,i_t})\}_{t=1}^T$ be the corresponding selected context–reward pairs. Given the gradient feature map $\mathbf{g}(\mathbf{x}; \theta_0) \in \mathbb{R}^p$, define for each $t \in [T]$,

$$\begin{aligned} \mathbf{V}_t &:= \mathbf{I} + \sum_{\tau=1}^t \mathbf{g}(\mathbf{x}_{\tau,i_\tau}; \theta_0) \mathbf{g}(\mathbf{x}_{\tau,i_\tau}; \theta_0)^\top \in \mathbb{R}^{p \times p}, \\ \mathbf{u}_t &:= \sum_{\tau=1}^t r_{\tau,i_\tau} \mathbf{g}(\mathbf{x}_{\tau,i_\tau}; \theta_0) \in \mathbb{R}^p, \\ \hat{\theta}_t &:= \mathbf{V}_t^{-1} \mathbf{u}_t \in \mathbb{R}^p. \end{aligned}$$

Lemma D.14 (Spectral and log-determinant bounds for $\mathbf{V}_T$). Let $\{i_t\}_{t=1}^T$ be the selected-arm sequence, and define $\mathbf{V}_T$ based on the selected contexts $\{\mathbf{x}_{t,i_t}\}_{t=1}^T$ as in Definition D.13. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random initialization θ_0 , the following hold:

$$(a) \|\mathbf{V}_T\|_2 \leq 1 + \mathcal{O}(TL);$$

$$(b) \log \det(\mathbf{V}_T) \leq \zeta \log(1 + T) + 1.$$

Proof. For (a), by Definition D.13 we have

$$\mathbf{V}_T = \mathbf{I} + \sum_{t=1}^T \mathbf{g}(\mathbf{x}_{t,i_t}; \theta_0) \mathbf{g}(\mathbf{x}_{t,i_t}; \theta_0)^\top.$$

Using $\|\mathbf{I}\|_2 = 1$ and $\|\mathbf{a}\mathbf{a}^\top\|_2 = \|\mathbf{a}\|_2^2$, it follows that

$$\|\mathbf{V}_T\|_2 \leq 1 + \sum_{t=1}^T \|\mathbf{g}(\mathbf{x}_{t,i_t}; \theta_0)\|_2^2 \leq 1 + \mathcal{O}(TL),$$

where the last inequality uses Lemma D.1 (b).

For (b), we have

$$\mathbf{G}^{(0)} := [\mathbf{g}(\mathbf{x}_{1,i_1}; \theta_0), \dots, \mathbf{g}(\mathbf{x}_{T,i_T}; \theta_0)] \in \mathbb{R}^{p \times T}.$$

Then $\mathbf{V}_T = \mathbf{I} + \mathbf{G}^{(0)}(\mathbf{G}^{(0)})^\top$, and hence

$$\log \det(\mathbf{V}_T) = \log \det(\mathbf{I} + \mathbf{G}^{(0)}(\mathbf{G}^{(0)})^\top).$$

By Sylvester's determinant identity $\det(\mathbf{I} + AB) = \det(\mathbf{I} + BA)$,

$$\log \det(\mathbf{I} + \mathbf{G}^{(0)}(\mathbf{G}^{(0)})^\top) = \log \det(\mathbf{I} + (\mathbf{G}^{(0)})^\top \mathbf{G}^{(0)}).$$

Let $\mathbf{H} \in \mathbb{R}^{T^k \times T^k}$ be the induced NTK matrix defined in Definition C.1 over the flattened list $\{\mathbf{x}_s\}_{s=1}^{T^k}$, where each s corresponds to a unique pair $(t, i) \in [T] \times [k]$. Given the selected-arm sequence $\{i_t\}_{t=1}^T$, let $s_t \in [T^k]$ denote the flattened index corresponding to (t, i_t) . We define $\mathbf{H}_T \in \mathbb{R}^{T \times T}$ as the principal submatrix of $\mathbf{H}$ indexed by $\{s_t\}_{t=1}^T$, i.e.,

$$(\mathbf{H}_T)_{t,\tau} := \mathbf{H}_{s_t, s_\tau}, \quad t, \tau \in [T].$$

Using the concavity of $\log \det(\cdot)$ and writing $(\mathbf{G}^{(0)})^\top \mathbf{G}^{(0)} = \mathbf{H}_T + ((\mathbf{G}^{(0)})^\top \mathbf{G}^{(0)} - \mathbf{H}_T)$, we obtain

$$\begin{aligned} \log \det(\mathbf{I} + (\mathbf{G}^{(0)})^\top \mathbf{G}^{(0)}) &\leq \log \det(\mathbf{I} + \mathbf{H}_T) + \left\langle (\mathbf{I} + \mathbf{H}_T)^{-1}, (\mathbf{G}^{(0)})^\top \mathbf{G}^{(0)} - \mathbf{H}_T \right\rangle \\ &\leq \log \det(\mathbf{I} + \mathbf{H}_T) + \|(\mathbf{I} + \mathbf{H}_T)^{-1}\|_F \cdot \|(\mathbf{G}^{(0)})^\top \mathbf{G}^{(0)} - \mathbf{H}_T\|_F \\ &\leq \log \det(\mathbf{I} + \mathbf{H}_T) + \sqrt{T} \|(\mathbf{G}^{(0)})^\top \mathbf{G}^{(0)} - \mathbf{H}_T\|_F, \end{aligned}$$

where we used $\|(\mathbf{I} + \mathbf{H}_T)^{-1}\|_F \leq \|\mathbf{I}^{-1}\|_F = \sqrt{T}$. By Lemma D.10 (applied to the T selected contexts) and choosing m sufficiently large, we have $\|(\mathbf{G}^{(0)})^\top \mathbf{G}^{(0)} - \mathbf{H}_T\|_F \leq 1/\sqrt{T}$ with probability at least $1 - \delta$. Therefore,

$$\log \det(\mathbf{V}_T) \leq \log \det(\mathbf{I} + \mathbf{H}_T) + 1 \leq \zeta \log(1 + T) + 1,$$

where the last inequality follows from the definition of ζ under the same convention as in the Theorem 20. This concludes the proof. $\square$

Lemma D.15 (Pointwise prediction error bound). *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random initialization θ_0 , there exists $\theta' \in \mathcal{B}(\theta_0, \tilde{\Omega}(T^{3/2}))$ such that the following holds simultaneously for all $(t, i) \in [T] \times [k]$:*

$$|y(\mathbf{x}_{t,i}) - f(\mathbf{x}_{t,i}; \theta')| \leq \mathcal{O}\left(\sqrt{\log \det(\mathbf{V}_t) - 2 \log \delta} + \sqrt{2S} + 1\right) \|\mathbf{g}(\mathbf{x}_{t,i}; \theta_0)\|_{\mathbf{V}_t^{-1}} + \mathcal{O}\left(\frac{T^2 L^3 \sqrt{\log m}}{m^{1/3}}\right). \quad (84)$$

Proof. Fix any $(t, i) \in [T] \times [k]$ and write $x := \mathbf{x}_{t,i}$ for brevity. Starting from the triangle inequality, we decompose

$$|y(x) - f(x; \theta')| \leq |y(x) - \langle \mathbf{g}(x; \theta_0), \hat{\theta}_t \rangle| + |f(x; \theta') - \langle \mathbf{g}(x; \theta_0), \hat{\theta}_t \rangle|. \quad (85)$$

We first control the term $|y(x) - \langle \mathbf{g}(x; \theta_0), \hat{\theta}_t \rangle|$. By Lemma D.11, on an event of probability at least $1 - \delta$, there exists $\theta^* \in \mathbb{R}^p$ such that, simultaneously for all (t, i) ,

$$y(\mathbf{x}_{t,i}) = \langle \mathbf{g}(\mathbf{x}_{t,i}; \theta_0), \theta^* - \theta_0 \rangle, \quad \text{and} \quad \|\theta^* - \theta_0\|_2 \leq \sqrt{2S}.$$

Therefore,

$$\begin{aligned} |y(x) - \langle \mathbf{g}(x; \theta_0), \hat{\theta}_t \rangle| &= |\langle \mathbf{g}(x; \theta_0), (\theta^* - \theta_0) - \hat{\theta}_t \rangle| \\ &\leq \|\mathbf{g}(x; \theta_0)\|_{\mathbf{V}_t^{-1}} \|(\theta^* - \theta_0) - \hat{\theta}_t\|_{\mathbf{V}_t}, \end{aligned} \quad (86)$$

where we used Cauchy–Schwarz in the $\mathbf{V}_t$ -geometry. Applying Theorem 2 of Abbasi-Yadkori et al. (2011b) with regularization $\lambda = 1$ (and identifying $\mathbf{V}_t$ with their design matrix) yields that, with probability at least $1 - \delta$, simultaneously for all t ,

$$\|(\theta^* - \theta_0) - \hat{\theta}_t\|_{\mathbf{V}_t} \leq \mathcal{O}\left(\sqrt{\log \det(\mathbf{V}_t) - 2 \log \delta} + \sqrt{2S}\right). \quad (87)$$

Combining (86) and (87) gives

$$|y(x) - \langle \mathbf{g}(x; \theta_0), \hat{\theta}_t \rangle| \leq \mathcal{O}\left(\sqrt{\log \det(\mathbf{V}_t) - 2 \log \delta} + \sqrt{2S}\right) \|\mathbf{g}(x; \theta_0)\|_{\mathbf{V}_t^{-1}}. \quad (88)$$

We next bound $|f(x; \theta') - \langle \mathbf{g}(x; \theta_0), \hat{\theta}_t \rangle|$ by adding and subtracting $\langle \mathbf{g}(x; \theta_0), \theta' - \theta_0 \rangle$:

$$|f(x; \theta') - \langle \mathbf{g}(x; \theta_0), \hat{\theta}_t \rangle| \leq |f(x; \theta') - \langle \mathbf{g}(x; \theta_0), \theta' - \theta_0 \rangle| + |\langle \mathbf{g}(x; \theta_0), (\theta' - \theta_0) - \hat{\theta}_t \rangle|. \quad (89)$$

For the first term on the right-hand side, using $f(x; \theta_0) = 0$ at initialization and Lemma D.2 (second-order remainder of linearization), we have

$$|f(x; \theta') - \langle \mathbf{g}(x; \theta_0), \theta' - \theta_0 \rangle| = |f(x; \theta') - f(x; \theta_0) - \langle \mathbf{g}(x; \theta_0), \theta' - \theta_0 \rangle| \leq \mathcal{O}(\omega^{4/3} L^3 \sqrt{\log m}), \quad (90)$$

where $\theta' \in \mathcal{B}(\theta_0, \omega m^{1/4})$. With $\omega = \tilde{\mathcal{O}}(T^{3/2})/m^{1/4}$, the right-hand side becomes $\mathcal{O}\left(\frac{T^2 L^3 \sqrt{\log m}}{m^{1/3}}\right)$.

For the second term in (89), another Cauchy–Schwarz inequality yields

$$\begin{aligned} |\langle \mathbf{g}(x; \theta_0), (\theta' - \theta_0) - \hat{\theta}_t \rangle| &\leq \|\mathbf{g}(x; \theta_0)\|_{\mathbf{V}_t^{-1}} \|(\theta' - \theta_0) - \hat{\theta}_t\|_{\mathbf{V}_t} \\ &\leq \|\mathbf{g}(x; \theta_0)\|_{\mathbf{V}_t^{-1}} \sqrt{\|\mathbf{V}_t\|_2} \|(\theta' - \theta_0) - \hat{\theta}_t\|_2. \end{aligned} \quad (91)$$

By Lemma D.14(a), $\|\mathbf{V}_t\|_2 \leq 1 + \mathcal{O}(tL)$, and by Lemma D.12(b), we can choose θ' so that $\|(\theta' - \theta_0) - \hat{\theta}_t\|_2 \leq 1/(1 + \mathcal{O}(tL))$. Plugging these into (91) gives

$$|\langle \mathbf{g}(x; \theta_0), (\theta' - \theta_0) - \hat{\theta}_t \rangle| \leq \mathcal{O}(1) \|\mathbf{g}(x; \theta_0)\|_{\mathbf{V}_t^{-1}}. \quad (92)$$

Substituting (88), (90), and (92) back into (85) yields

$$|y(\mathbf{x}_{t,i}) - f(\mathbf{x}_{t,i}; \theta')| \leq \mathcal{O}\left(\sqrt{\log \det(\mathbf{V}_t) - 2 \log \delta} + \sqrt{2S} + 1\right) \|\mathbf{g}(\mathbf{x}_{t,i}; \theta_0)\|_{\mathbf{V}_t^{-1}} + \mathcal{O}\left(\frac{T^2 L^3 \sqrt{\log m}}{m^{1/3}}\right),$$

which completes the proof. $\square$

Lemma D.16 (Global squared error bound along the selected trajectory). *Let $\{i_t\}_{t=1}^T$ be the selected-arm sequence, and let $\{(\mathbf{x}_{t,i_t}, r_{t,i_t})\}_{t=1}^T$ be the corresponding selected context–reward pairs. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random initialization θ_0 , there exists a parameter $\theta' \in \mathcal{B}(\theta_0, \tilde{\Omega}(T^{3/2}))$ such that*

$$\mathbb{E} \left[\sum_{t=1}^T \frac{1}{2} (r_{t,i_t} - f(\mathbf{x}_{t,i_t}; \theta'))^2 \right] \leq \mathcal{O}\left(\sqrt{\zeta \log(1+T) - 2 \log \delta} + \sqrt{2S} + 1\right)^2 \cdot \zeta \log(1+T), \quad (93)$$

where the expectation is taken with respect to the reward noise.

Proof. Let $\{i_t\}_{t=1}^T$ be the selected-arm sequence and write $\mathbf{x}_t := \mathbf{x}_{t,i_t}$, $r_t := r_{t,i_t}$ for brevity. Recall that $\mathbb{E}[r_t \mid \mathbf{x}_t] = y(\mathbf{x}_t)$. Therefore,

$$\sum_{t=1}^T \mathbb{E}[(r_t - f(\mathbf{x}_t; \theta'))^2] = \sum_{t=1}^T (y(\mathbf{x}_t) - f(\mathbf{x}_t; \theta'))^2. \quad (94)$$

We upper bound each squared error term using Lemma D.15. For every $t \in [T]$, that lemma gives (on the same high-probability event)

$$|y(\mathbf{x}_t) - f(\mathbf{x}_t; \theta')| \leq \mathcal{O}\left(\sqrt{\log \det(\mathbf{V}_t)} - 2 \log \delta + \sqrt{2}S + 1\right) \|g(\mathbf{x}_t; \theta_0)\|_{\mathbf{V}_t^{-1}} + \mathcal{O}\left(\frac{T^2 L^3 \sqrt{\log m}}{m^{1/3}}\right).$$

Applying $(a + b)^2 \leq 2a^2 + 2b^2$ and summing over t yields

$$\begin{aligned} \sum_{t=1}^T (y(\mathbf{x}_t) - f(\mathbf{x}_t; \theta'))^2 &\leq \mathcal{O}\left(\sqrt{\log \det(\mathbf{V}_T)} - 2 \log \delta + \sqrt{2}S + 1\right)^2 \sum_{t=1}^T \|g(\mathbf{x}_t; \theta_0)\|_{\mathbf{V}_t^{-1}}^2 \\ &\quad + 2T \cdot \mathcal{O}\left(\frac{T^2 L^3 \sqrt{\log m}}{m^{1/3}}\right)^2, \end{aligned} \quad (95)$$

where we used the monotonicity $\log \det(\mathbf{V}_t) \leq \log \det(\mathbf{V}_T)$.

It remains to control the self-normalized sum in (95). By Lemma 11 of Abbasi-Yadkori et al. (2011b) (with $\mathbf{V}_t = \mathbf{I} + \sum_{\tau=1}^t g(\mathbf{x}_\tau; \theta_0)g(\mathbf{x}_\tau; \theta_0)^\top$),

$$\sum_{t=1}^T \|g(\mathbf{x}_t; \theta_0)\|_{\mathbf{V}_t^{-1}}^2 \leq 2 \log \det(\mathbf{V}_T). \quad (96)$$

Combining (96) with Lemma D.14(b), i.e., $\log \det(\mathbf{V}_T) \leq \zeta \log(1 + T) + 1$, we obtain

$$\sum_{t=1}^T \|g(\mathbf{x}_t; \theta_0)\|_{\mathbf{V}_t^{-1}}^2 \leq 2(\zeta \log(1 + T) + 1).$$

Plugging this back into (95) and using the width condition to absorb $2T \cdot \mathcal{O}\left(\frac{T^2 L^3 \sqrt{\log m}}{m^{1/3}}\right)^2$ into $\mathcal{O}(1)$, we have

$$\sum_{t=1}^T (y(\mathbf{x}_t) - f(\mathbf{x}_t; \theta'))^2 \leq \mathcal{O}\left(\sqrt{\zeta \log(1 + T)} - 2 \log \delta + \sqrt{2}S + 1\right)^2 \cdot (\zeta \log(1 + T) + 1) + \mathcal{O}(1).$$

Finally, dividing both sides by 2 and using (94) gives (93). $\square$

D.4. Putting everything together

Proof of Theorem 4.4. Fix any round $t \in [T]$ and abbreviate $\hat{i} = \hat{i}_t$ and $i^* = i_t^*$. Recall that $\hat{i}$ maximizes the computed propagated score $\hat{\pi}_t[v_{t,i}]$ over $i \in [k]$, while i^* maximizes the ideal score $\pi_t^*[v_{t,i}]$.

We begin with a decomposition that inserts the exact PPR vector π_t (personalization $\mathbf{s}_t$) and the approximate output $\hat{\pi}_t$:

$$\begin{aligned} \pi_t^*[v_{t,i^*}] - \pi_t^*[v_{t,\hat{i}}] &= (\pi_t^*[v_{t,i^*}] - \pi_t[v_{t,i^*}]) + (\pi_t[v_{t,i^*}] - \pi_t[v_{t,\hat{i}}]) + (\pi_t[v_{t,\hat{i}}] - \pi_t^*[v_{t,\hat{i}}]) \\ &\leq |\pi_t^*[v_{t,i^*}] - \pi_t[v_{t,i^*}]| + |\pi_t^*[v_{t,\hat{i}}] - \pi_t[v_{t,\hat{i}}]| + (\pi_t[v_{t,i^*}] - \pi_t[v_{t,\hat{i}}]). \end{aligned} \quad (97)$$

For the middle gap, insert $\hat{\pi}_t$ and use the choice of $\hat{i}$:

$$\begin{aligned} \pi_t[v_{t,i^*}] - \pi_t[v_{t,\hat{i}}] &= (\pi_t[v_{t,i^*}] - \hat{\pi}_t[v_{t,i^*}]) + (\hat{\pi}_t[v_{t,i^*}] - \hat{\pi}_t[v_{t,\hat{i}}]) + (\hat{\pi}_t[v_{t,\hat{i}}] - \pi_t[v_{t,\hat{i}}]) \\ &\leq |\pi_t[v_{t,i^*}] - \hat{\pi}_t[v_{t,i^*}]| + |\pi_t[v_{t,\hat{i}}] - \hat{\pi}_t[v_{t,\hat{i}}]|. \end{aligned} \quad (98)$$

Combining (97)–(98) yields

$$\pi_t^*[v_{t,i^*}] - \pi_t^*[v_{t,\hat{i}}] \leq 2\|\pi_t^* - \pi_t\|_\infty + \max_{i \in [k]} |\hat{\pi}_t[v_{t,i}] - \pi_t[v_{t,i}]| + \max_{i \in [k]} |\hat{\pi}_t[v_{t,i}] - \pi_t[v_{t,i}]|. \quad (99)$$

Define $\Delta_t := \max_{i \in [k]} |\hat{\pi}_t[v_{t,i}] - \pi_t[v_{t,i}]|$ and use $\|x\|_\infty \leq \|x\|_2$, we obtain the clean bound

$$\pi_t^*[v_{t,i^*}] - \pi_t^*[v_{t,\hat{i}}] \leq 2\|\pi_t^* - \pi_t\|_2 + 2\Delta_t. \quad (100)$$

Next we control the two terms on the right-hand side. Both π_t^* and π_t solve the PageRank fixed-point equation on the same graph $\mathcal{G}_{t-1}$ (with the same transition matrix $\mathbf{P}_{t-1}$), but with different personalization vectors $\mathbf{y}_t$ and $\mathbf{s}_t$. Using the closed form (Lemma E.14 / PageRank resolvent form),

$$\pi_t^* = (1 - \alpha)(\mathbf{I} - \alpha\mathbf{P}_{t-1})^{-1}\mathbf{y}_t, \quad \pi_t = (1 - \alpha)(\mathbf{I} - \alpha\mathbf{P}_{t-1})^{-1}\mathbf{s}_t.$$

Therefore,

$$\pi_t^* - \pi_t = (1 - \alpha)(\mathbf{I} - \alpha\mathbf{P}_{t-1})^{-1}(\mathbf{y}_t - \mathbf{s}_t).$$

Since $\mathbf{P}_{t-1}$ is row-stochastic, its spectral radius is at most 1, hence $\mathbf{I} - \alpha\mathbf{P}_{t-1} \succeq (1 - \alpha)\mathbf{I}$ and thus $\|(1 - \alpha)(\mathbf{I} - \alpha\mathbf{P}_{t-1})^{-1}\|_2 \leq 1$. Consequently,

$$\|\pi_t^* - \pi_t\|_2 \leq \|\mathbf{y}_t - \mathbf{s}_t\|_2. \quad (101)$$

It remains to bound Δ_t using the APPR guarantee (4). By definition of $\|\cdot\|_{\infty, D_{t-1}}$,

$$\|(\hat{\pi}_t - \pi_t)\|_{\infty, D_{t-1}} = \max_{v \in \mathcal{V}} \frac{|\hat{\pi}_t(v) - \pi_t(v)|}{d_v^{(t-1)}}.$$

Thus, from $\hat{\pi}_t \in \mathcal{P}_t(\epsilon, \alpha, \mathbf{s}_t, \mathcal{G}_{t-1})$ we get that for every node $v \in \mathcal{V}$,

$$|\hat{\pi}_t(v) - \pi_t(v)| \leq \epsilon d_v^{(t-1)}.$$

In particular, for each candidate node $v_{t,i}$ we have $|\hat{\pi}_t[v_{t,i}] - \pi_t[v_{t,i}]| \leq \epsilon d_{v_{t,i}}^{(t-1)} \leq \epsilon d_{\max}$, and hence

$$\Delta_t = \max_{i \in [k]} |\hat{\pi}_t[v_{t,i}] - \pi_t[v_{t,i}]| \leq \epsilon d_{\max}. \quad (102)$$

Plugging (101) and (102) into (100) and summing over t gives

$$\mathbf{R}_T \leq 2 \sum_{t=1}^T \|\mathbf{y}_t - \mathbf{s}_t\|_2 + 2\epsilon d_{\max} T. \quad (103)$$

We now control the first sum by Cauchy–Schwarz:

$$\sum_{t=1}^T \|\mathbf{y}_t - \mathbf{s}_t\|_2 \leq \sqrt{T} \left(\sum_{t=1}^T \|\mathbf{y}_t - \mathbf{s}_t\|_2^2 \right)^{1/2}.$$

By the definitions of $\mathbf{y}_t$ and $\mathbf{s}_t$ (stacking over $i \in [k]$),

$$\|\mathbf{y}_t - \mathbf{s}_t\|_2^2 = \sum_{i=1}^k (y(\mathbf{x}_{t,i}) - (\mu_{t,i} + b_{t,i}))^2.$$

Invoking Lemma D.16 (with the same θ' guaranteed there) and using $\mathbb{E}[r_{t,i} | \mathbf{x}_{t,i}] = y(\mathbf{x}_{t,i})$, we obtain

$$\begin{aligned} \sum_{t=1}^T \|\mathbf{y}_t - \mathbf{s}_t\|_2^2 &= \sum_{t=1}^T \sum_{i=1}^k (y(\mathbf{x}_{t,i}) - f(\mathbf{x}_{t,i}; \theta'))^2 = \sum_{t=1}^T \sum_{i=1}^k \mathbb{E}[(r_{t,i} - f(\mathbf{x}_{t,i}; \theta'))^2] \\ &\leq \tilde{O}(\zeta k \max(\zeta, S^2)), \end{aligned} \quad (104)$$

where we used $\log(1 + Tk)$ -type factors are absorbed into $\tilde{O}(\cdot)$ and the same ζ, S as in the theorem. Therefore,

$$\sum_{t=1}^T \|\mathbf{y}_t - \mathbf{s}_t\|_2 \leq \tilde{O}\left(\sqrt{T} \sqrt{\zeta k \max(\zeta, S^2)}\right). \quad (105)$$

Substituting (105) into (103) yields

$$\mathbf{R}_T \leq \tilde{\mathcal{O}}\left(\sqrt{T} \sqrt{\zeta k \max(\zeta, S^2)}\right) + 2\epsilon d_{\max} T.$$

Finally, setting $\epsilon = \mathcal{O}(1/\sqrt{T})$ gives $2\epsilon d_{\max} T = 2d_{\max} \sqrt{T}$, and hence

$$\mathbf{R}_T \leq \tilde{\mathcal{O}}\left(\sqrt{T} \left(d_{\max} + \sqrt{\zeta k \max(\zeta, S^2)}\right)\right),$$

which is exactly (20). This completes the proof. $\square$

E. Additional Experiments and Implementation Details

This appendix supplements the main paper with dataset statistics, detailed experimental setups for *online* link prediction, computational cost analysis, and additional ablation studies for LOCPRB.

E.1. Graph Dataset Statistics

We use (i) recommendation datasets: MovieLens (Harper & Konstan, 2015) and AmazonFashion (Ni et al., 2019), (ii) social network datasets: Facebook (Leskovec & McAuley, 2012) and GrQc (Leskovec et al., 2007), and (iii) large-scale OGB link prediction datasets: ogbl-Collab, ogbl-PPA, and ogbl-Vessel from the Open Graph Benchmark (Hu et al., 2020). Compared to the main text (Section 5.1), we additionally include AmazonFashion, Facebook, and GrQc to further evaluate robustness across diverse graph domains. Table 6 summarizes key dataset statistics.

Table 6. Graph dataset statistics.

Dataset	#Nodes	#Edges	Type	Context dim. (d)
MovieLens	12,000	508,572	Bipartite ($U \cup I$)	20
AmazonFashion	8,000	3,679	Bipartite ($U \cup I$)	20
Facebook	4,039	88,234	Unipartite	20
GrQc	5,242	26,082	Unipartite	20
ogbl-Collab	235,868	1,285,465	Unipartite	128
ogbl-PPA	576,289	30,326,273	Unipartite	58
ogbl-Vessel	3,538,495	5,345,897	Unipartite	3

E.2. Experiment Setups and Implementation Details

We follow the online link prediction protocol described in the main text and evaluate PRB and LOCPRB under the same data stream and candidate construction. For MovieLens/AmazonFashion, the node set is bipartite with users U and items I (i.e., $\mathcal{V} = U \cup I$). At each round we draw a user $v_t \in U$ and form a candidate pool $\mathcal{V}_t \subseteq I$, where positives are items purchased by v_t . For Facebook/GrQc, the node set is unipartite; we draw $v_t \in \mathcal{V}$ and form $\mathcal{V}_t \subseteq \mathcal{V}$, where positives are true neighbors of v_t in the original graph (their edges remain hidden until selected).

We start from a cold graph $\mathcal{G}_0 = (\mathcal{V}, \mathcal{E}_0 = \emptyset)$. We run for $T = 10,000$ rounds on MovieLens, Facebook, and GrQc, and for $T = 5,000$ rounds on AmazonFashion and the three OGB datasets. At each round, the candidate pool has a fixed size 10 and contains one ground-truth positive and nine uniformly sampled negatives, consistent with the main text. The learner selects $\hat{v}_t \in \mathcal{V}_t$, observes binary feedback, and updates the evolving graph by adding $(v_t, \hat{v}_t)$ iff the reward is 1 (otherwise $\mathcal{G}_t = \mathcal{G}_{t-1}$).

Both PRB and LOCPRB use two 2-layer fully-connected ReLU networks: an exploitation model $f_1(\cdot; \theta^1)$ and an exploration model $f_2(\cdot; \theta^2)$, each with hidden width 100. We use the random Gaussian initialization in Eq. (19). We optimize f_1 with SGD and optimize f_2 with Adam. For each dataset, we tune the learning rate and select the best-performing value; this choice is then fixed for the entire run, and PRB and LOCPRB use the same learning rate for fair comparison.

Following common practice in neural contextual bandits, we update network parameters every 50 rounds for the first 2,000 rounds, and every 100 rounds thereafter.

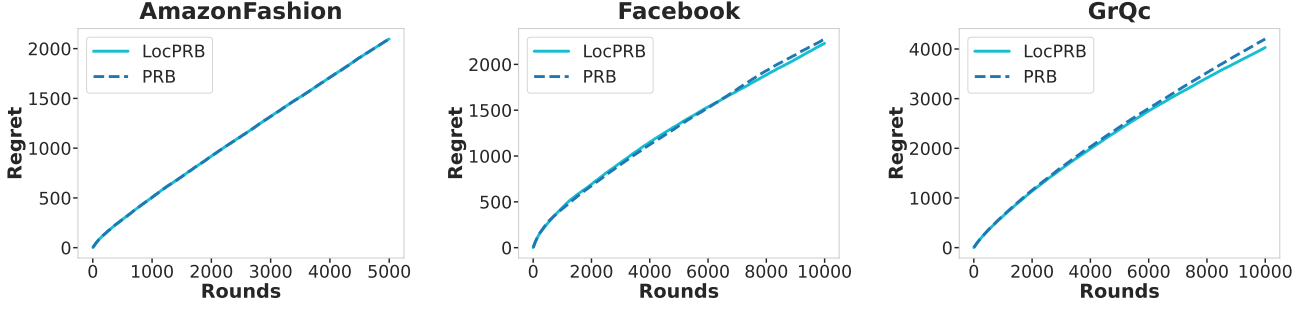


Figure 3. Regret comparison of PRB and LOCPRB on AmazonFashion, Facebook and GrQc (average of 10 runs).

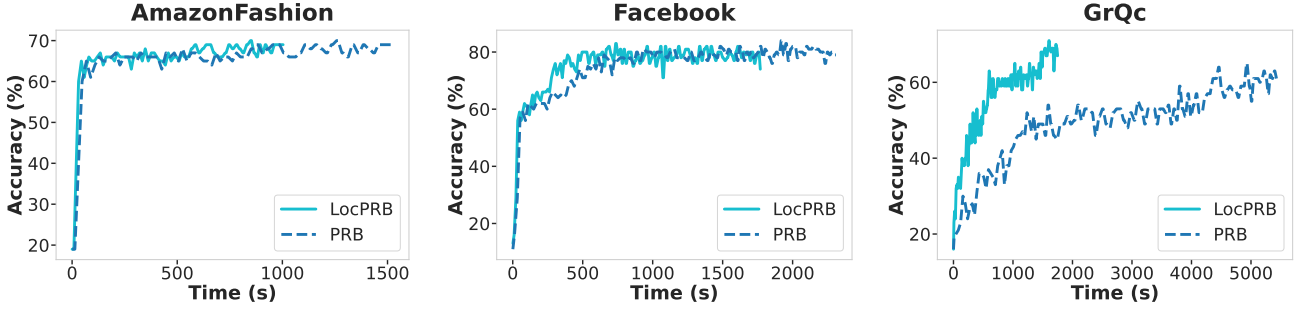


Figure 4. Test accuracy versus time for PRB and LOCPRB on AmazonFashion, Facebook and GrQc, evaluated periodically at fixed checkpoints during online learning (average of 10 runs).

For propagation, PRB uses PageRank damping $\alpha = 0.85$ and computes the PageRank vector via power iteration for 50 iterations. LOCPRB uses the same damping factor $\alpha = 0.85$ and computes ϵ -approximate PPR using APPR, with tolerance set to $\epsilon = 1/n$ by default (and varied only in ablations). To accelerate across rounds, LOCPRB employs DYN-APPR: we warm-start from the previous $(\mathbf{p}, \mathbf{r})$ state, apply INSERTUPDATE after each edge insertion, and then run local pushes until the stopping condition is satisfied.

We report the regret trajectory over rounds, the cumulative regret at the final horizon T , and the total running time. To reduce randomness, we repeat each experiment with 10 random seeds and report mean $\pm$ std for both cumulative regret and running time.

E.3. Additional Results on AmazonFashion, Facebook, and GrQc

To further assess robustness across graph domains beyond the main benchmarks, we additionally evaluate PRB and LOCPRB on AmazonFashion (recommendation), Facebook (social network), and GrQc (collaboration network). Figure 3 reports the cumulative regret trajectories over rounds, and Figure 4 plots test accuracy as a function of wall-clock time, where accuracy is measured periodically at fixed evaluation checkpoints during online learning (both averaged over 10 runs).

As shown in Figure 3, LOCPRB closely matches the regret trajectory of PRB across all three datasets, indicating that replacing full propagation with warm-started local pushes preserves the decision quality of PageRank-based scoring. Notably, on GrQc, LOCPRB achieves a small but consistent improvement, suggesting that localized diffusion can better capture the relevant neighborhood signals in this domain.

Figure 4 highlights the accuracy–efficiency tradeoff in wall-clock time. On AmazonFashion and Facebook, LOCPRB provides accuracy that is comparable to PRB, with only minor differences throughout training. In contrast, on GrQc, LOCPRB reaches higher accuracy faster and maintains a clear advantage over time, demonstrating that dynamic local propagation can yield substantial practical gains when full diffusion is more costly or less well-aligned with the task’s effective neighborhood scale.

F. Limitations.

LOCPRB focuses on coupling neural exploitation–exploration with Personalized PageRank, and we do not investigate other ways to integrate bandit scoring with graph structure, such as alternative random-walk diffusions (e.g., heat-kernel) or GNN-based propagation modules. Our evaluation is centered on online link prediction; while the same bandit–propagation template is broadly applicable, other dynamic graph tasks (e.g., node classification, clustering, and subgraph matching) are not explored here. Extending LOCPRB to these operators and tasks, and studying how different propagation choices affect both decision quality and efficiency, is an important direction for future work.